ExamGOAL
PYQ PDF

Free · With answer key

JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Binomial Theorem

JEE Main Mathematics · Previous Year Questions with Answer Key

271	193	2002 – 2026	75	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

MCQ (Single Correct Answer)	127 questions
Numerical	69 questions
Out of syllabus	75 questions
Answer key	at the end

Practise online

Every question with its full solution, free on **questions.examside.com**

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

MCQ (Single Correct Answer) 127 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

If $26\left(\frac{2^3}{3}\left({}^{12}C_2\right)+\frac{2^5}{5}\left({}^{12}C_4\right)+\frac{2^7}{7}\left({}^{12}C_6\right)+\cdots+\frac{2^{13}}{13}\left({}^{12}C_{12}\right)\right)=3^{13}-\alpha$, then α is equal to :

(A) 45 (B) 48 (C) 51 (D) 54

Open solution

Q2. JEE Main 2026 (Online) 6th April Morning Shift

If the coefficients of the middle terms in the binomial expansions of $(1+\alpha x)^{26}$ and $(1-\alpha x)^{28}, \alpha \neq 0$, are equal, then the value of α is:

(A) 1 (B) $\frac{14}{13}$ (C) $\frac{27}{7}$ (D) $\frac{7}{27}$

Open solution

Q3. JEE Main 2026 (Online) 5th April Evening Shift

The coefficient of x^2 in the expansion of $\left(2x^2+\frac{1}{x}\right)^{10}, x \neq 0$, is :

(A) 3240 (B) 3360 (C) 3480 (D) 3600

Open solution

Q4. JEE Main 2026 (Online) 4th April Evening Shift

In the expansion of $\left(9x-\frac{1}{3\sqrt{x}}\right)^{18}, x>0$, if the term independent of x is (221)k, then k is equal to:

(A) 84 (B) 78 (C) 168 (D) 198

Open solution

Q5. JEE Main 2026 (Online) 4th April Morning Shift

Let the smallest value of $k \in \mathbb{N}$, for which the coefficient of x^3 in $(1+x)^3+(1+x)^4+(1+x)^5+\ldots+(1+x)^{99}+(1+kx)^{100}, x \neq 0$, is $\left(43n+\frac{101}{4}\right)\left({}^{100}C_3\right)$ for some $n \in \mathbb{N}$, be p. Then the value of $p+n$ is :

(A) 10 (B) 11 (C) 12 (D) 13

Open solution

Q6. JEE Main 2026 (Online) 2nd April Evening Shift

If for $3 \leq r \leq 30$, $\left({}^{30}C_{30-r}\right)+3\left({}^{30}C_{31-r}\right)+3\left({}^{30}C_{32-r}\right)+\left({}^{30}C_{33-r}\right)={}^{m}C_r$, then m equals :

(A) 31 (B) 32 (C) 33 (D) 34

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q7. JEE Main 2026 (Online) 28th January Evening Shift

Open solution

Given below are two statements :

Statement I :

$25^{13} + 20^{13} + 8^{13} + 3^{13}$ is divisible by 7.

Statement II :

The integral part of $(7 + 4\sqrt{3})^{25}$ is an odd number.

In the light of the above statements, choose the **correct answer** from the options given below :

(A) **Statement I** is false but **Statement II** is true

(B) Both **Statement I** and **Statement II** are false

(C) Both **Statement I** and **Statement II** are true

(D) **Statement I** is true but **Statement II** is false

Q8. JEE Main 2026 (Online) 28th January Evening Shift

Open solution

The sum of the coefficients of x^{499} and x^{500} in $(1+x)^{1000} + x(1+x)^{999} + x^2(1+x)^{998} + \ldots + x^{1000}$ is :

(A) ${}^{1002}C_{501}$ (B) ${}^{1001}C_{501}$ (C) ${}^{1000}C_{501}$ (D) ${}^{1002}C_{500}$

Q9. JEE Main 2026 (Online) 24th January Morning Shift

Open solution

Let $S = \frac{1}{25!} + \frac{1}{3!23!} + \frac{1}{5!21!} + \ldots$ up to 13 terms. If $13\,S = \frac{2^k}{n!}, k \in N$, then $n + k$ is equal to

(A) 50 (B) 52 (C) 49 (D) 51

Q10. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

The sum of all possible values of $n \in N$, so that the coefficients of x, x^2 and x^3 in the expansion of $\left(1+x^2\right)^2(1+x)^n$, are in arithmetic progression is :

(A) 12 (B) 9 (C) 3 (D) 7

Q11. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

The value of $\frac{{}^{100}C_{50}}{51} + \frac{{}^{100}C_{51}}{52} + \ldots . + \frac{{}^{100}C_{100}}{101}$ is:

(A) $\frac{2^{101}}{101}$ (B) $\frac{2^{100}}{101}$ (C) $\frac{2^{100}}{100}$ (D) $\frac{2^{101}}{100}$

Q12. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

Let C_r denote the coefficient of x^r in the binomial expansion of $(1+x)^n, n \in N, 0 \le r \le n$. If

$P_n = C_0 - C_1 + \frac{2^2}{3}C_2 - \frac{2^3}{4}C_3 + \ldots . + \frac{(-2)^n}{n+1}C_n$, then the value of $\sum_{n=1}^{25} \frac{1}{P_{2n}}$ equals.

(A) 675 (B) 580 (C) 525 (D) 650

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q13. JEE Main 2026 (Online) 22nd January Morning Shift

The coefficient of x^{48} in $(1+x)+2(1+x)^2+3(1+x)^3+\ldots+100(1+x)^{100}$ is equal to

(A) $100 \cdot {}^{100}C_{49} - {}^{100}C_{48}$
(B) $100 \cdot {}^{101}C_{49} - {}^{101}C_{50}$
(C) ${}^{100}C_{50} + {}^{101}C_{49}$
(D) $100 \cdot {}^{100}C_{49} - {}^{100}C_{50}$

Open solution

Q14. JEE Main 2026 (Online) 21st January Morning Shift

If the coefficient of x in the expansion of $(ax^2+bx+c)(1-2x)^{26}$ is -56 and the coefficients of x^2 and x^3 are both zero, then $\mathrm{a}+\mathrm{b}+\mathrm{c}$ is equal to :

(A) 1483
(B) 1300
(C) 1500
(D) 1403

Open solution

Q15. JEE Main 2025 (Online) 8th April Evening Shift

The number of integral terms in the expansion of $\left(5^{\frac{1}{2}}+7^{\frac{1}{8}}\right)^{1016}$ is:

(A) 127
(B) 128
(C) 130
(D) 129

Open solution

Q16. JEE Main 2025 (Online) 7th April Morning Shift

The remainder when $\left((64)^{(64)}\right)^{(64)}$ is divided by 7 is equal to

(A) 4
(B) 6
(C) 3
(D) 1

Open solution

Q17. JEE Main 2025 (Online) 4th April Evening Shift

If $1^2 \cdot \left({}^{15}C_1\right) + 2^2 \cdot \left({}^{15}C_2\right) + 3^2 \cdot \left({}^{15}C_3\right) + \ldots + 15^2 \cdot \left({}^{15}C_{15}\right) = 2^m \cdot 3^n \cdot 5^k$, where $m, n, k \in \mathbf{N}$, then $\mathrm{m}+\mathrm{n}+\mathrm{k}$ is equal to :

(A) 20
(B) 19
(C) 18
(D) 21

Open solution

Q18. JEE Main 2025 (Online) 4th April Morning Shift

For an integer $n \geq 2$, if the arithmetic mean of all coefficients in the binomial expansion of $(x+y)^{2n-3}$ is 16 , then the distance of the point $\mathrm{P}\left(2n-1, n^2-4n\right)$ from the line $x+y=8$ is

(A) $\sqrt{2}$
(B) $2\sqrt{2}$
(C) $5\sqrt{2}$
(D) $3\sqrt{2}$

Open solution

Q19. JEE Main 2025 (Online) 4th April Morning Shift

In the expansion of $\left(\sqrt[3]{2}+\frac{1}{\sqrt[3]{3}}\right)^n, n \in \mathrm{N}$, if the ratio of 15th term from the beginning to the 15th term from the end is $\frac{1}{6}$, then the value of ${}^nC_3$ is

(A) 4960
(B) 2300
(C) 1040
(D) 4060

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q20. JEE Main 2025 (Online) 3rd April Morning Shift

The sum of all rational terms in the expansion of $(2+\sqrt{3})^8$ is :

(A) 16923 (B) 18817 (C) 3763 (D) 33845

Open solution

Q21. JEE Main 2025 (Online) 3rd April Morning Shift

If $\sum_{r=1}^{9}\left(\frac{r+3}{2^r}\right)\cdot {}^9C_r = \alpha\left(\frac{3}{2}\right)^9 - \beta, \alpha, \beta \in \mathbb{N}$, then $(\alpha+\beta)^2$ is equal to

(A) 27 (B) 81 (C) 18 (D) 9

Open solution

Q22. JEE Main 2025 (Online) 2nd April Evening Shift

$If \sum_{r=0}^{10}\left(\frac{10^{r+1}-1}{10^r}\right).{}^{11}C_{r+1} = \frac{\alpha^{11}-11^{11}}{10^{10}}, then\ \alpha\ is\ equal\ to:$

(A) 11 (B) 20 (C) 24 (D) 15

Open solution

Q23. JEE Main 2025 (Online) 2nd April Morning Shift

The largest $n \in \mathbf{N}$ such that 3^n divides 50 ! is :

(A) 22 (B) 20 (C) 21 (D) 23

Open solution

Q24. JEE Main 2025 (Online) 2nd April Morning Shift

The term independent of x in the expansion of $\left(\frac{(x+1)}{(x^{2/3}+1-x^{1/3})}-\frac{(x-1)}{(x-x^{1/2})}\right)^{10}, x>1$, is :

(A) 240 (B) 120 (C) 150 (D) 210

Open solution

Q25. JEE Main 2025 (Online) 29th January Evening Shift

The remainder, when 7^{103} is divided by 23, is equal to:

(A) 9 (B) 6 (C) 14 (D) 17

Open solution

Q26. JEE Main 2025 (Online) 29th January Morning Shift

The least value of **n** for which the number of integral terms in the Binomial expansion of $(\sqrt[3]{7}+\sqrt[12]{11})^n$ is 183, is :

(A) 2184 (B) 2172 (C) 2196 (D) 2148

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q27. JEE Main 2025 (Online) 28th January Evening Shift

Let the coefficients of three consecutive terms T_r, T_{r+1} and T_{r+2} in the binomial expansion of $(a+b)^{12}$ be in a G.P. and let p be the number of all possible values of r. Let q be the sum of all rational terms in the binomial expansion of $(\sqrt[4]{3}+\sqrt[3]{4})^{12}$. Then $p+q$ is equal to:

Open solution

(A) 295 (B) 283 (C) 299 (D) 287

Q28. JEE Main 2025 (Online) 24th January Evening Shift

Suppose A and B are the coefficients of 30th and 12th terms respectively in the binomial expansion of $(1+x)^{2n-1}$. If $2\,A = 5\,B$, then n is equal to:

Open solution

(A) 20 (B) 19 (C) 22 (D) 21

Q29. JEE Main 2025 (Online) 24th January Morning Shift

For some $n \neq 10$, let the coefficients of the 5 th, 6 th and 7 th terms in the binomial expansion of $(1+x)^{n+4}$ be in A.P. Then the largest coefficient in the expansion of $(1+x)^{n+4}$ is:

Open solution

(A) 10 (B) 35 (C) 70 (D) 20

Q30. JEE Main 2025 (Online) 23rd January Evening Shift

If in the expansion of $(1+x)^p(1-x)^q$, the coefficients of x and x^2 are 1 and -2 , respectively, then p^2+q^2 is equal to :

Open solution

(A) 8 (B) 20 (C) 13 (D) 18

Q31. JEE Main 2025 (Online) 22nd January Evening Shift

Let α, β, γ and δ be the coefficients of x^7, x^5, x^3 and x respectively in the expansion of $\left(x+\sqrt{x^3-1}\right)^5+\left(x-\sqrt{x^3-1}\right)^5, x>1$. If u and v satisfy the equations

Open solution

$\alpha u+\beta v=18,$

$\gamma u+\delta v=20,$

then $u+v$ equals :

(A) 4 (B) 3 (C) 5 (D) 8

Q32. JEE Main 2024 (Online) 9th April Evening Shift

The sum of the coefficients of $x^{2/3}$ and $x^{-2/5}$ in the binomial expansion of $\left(x^{2/3}+\frac{1}{2}x^{-2/5}\right)^9$ is

Open solution

(A) 19/4 (B) 69/16 (C) 63/16 (D) 21/4

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q33. JEE Main 2024 (Online) 9th April Morning Shift

The coefficient of x^{70} in $x^2(1+x)^{98}+x^3(1+x)^{97}+x^4(1+x)^{96}+\ldots+x^{54}(1+x)^{46}$ is ${}^{99}\mathrm{C}_{\mathrm{p}}-{}^{46}\mathrm{C}_{\mathrm{q}}$. Then a possible value of $\mathrm{p}+\mathrm{q}$ is :

(A) 61 (B) 83 (C) 55 (D) 68

Open solution

Q34. JEE Main 2024 (Online) 8th April Evening Shift

If the term independent of x in the expansion of $\left(\sqrt{\mathrm{a}}x^2+\frac{1}{2x^3}\right)^{10}$ is 105 , then a^2 is equal to :

(A) 6 (B) 4 (C) 2 (D) 9

Open solution

Q35. JEE Main 2024 (Online) 5th April Evening Shift

If the constant term in the expansion of $\left(\frac{\sqrt[5]{3}}{x}+\frac{2x}{\sqrt[3]{5}}\right)^{12}, x\neq 0$, is $\alpha\times 2^8\times\sqrt[5]{3}$, then 25α is equal to :

(A) 724 (B) 742 (C) 693 (D) 639

Open solution

Q36. JEE Main 2024 (Online) 4th April Evening Shift

If the coefficients of x^4, x^5 and x^6 in the expansion of $(1+x)^n$ are in the arithmetic progression, then the maximum value of n is:

(A) 28 (B) 21 (C) 7 (D) 14

Open solution

Q37. JEE Main 2024 (Online) 4th April Morning Shift

The sum of all rational terms in the expansion of $\left(2^{\frac{1}{5}}+5^{\frac{1}{3}}\right)^{15}$ is equal to :

(A) 633 (B) 6131 (C) 3133 (D) 931

Open solution

Q38. JEE Main 2024 (Online) 1st February Evening Shift

Let m and n be the coefficients of seventh and thirteenth terms respectively

in the expansion of $\left(\frac{1}{3}x^{\frac{1}{3}}+\frac{1}{2x^{\frac{2}{3}}}\right)^{18}$. Then $\left(\frac{\mathrm{n}}{\mathrm{m}}\right)^{\frac{1}{3}}$ is :

(A) $\frac{1}{9}$ (B) $\frac{1}{4}$ (C) $\frac{4}{9}$ (D) $\frac{9}{4}$

Open solution

Q39. JEE Main 2024 (Online) 31st January Morning Shift

Let a be the sum of all coefficients in the expansion of $\left(1-2x+2x^2\right)^{2023}\left(3-4x^2+2x^3\right)^{2024}$ and $b=\lim\limits_{x\to 0}\left(\frac{\int_0^x\frac{\log(1+t)}{t^{2024}+1}dt}{x^2}\right)$. If the equation $cx^2+dx+e=0$ and $2bx^2+ax+4=0$ have a common root, where $c,d,e\in\mathbb{R}$, then $\mathrm{d}:\mathrm{c}:\mathrm{e}$ equals

(A) $2:1:4$ (B) $1:1:4$ (C) $1:2:4$ (D) $4:1:4$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q40. JEE Main 2024 (Online) 30th January Evening Shift

Suppose $2-p, p, 2-\alpha, \alpha$ are the coefficients of four consecutive terms in the expansion of $(1+x)^n$. Then the value of $p^2-\alpha^2+6\alpha+2p$ equals

(A) 8 (B) 4 (C) 6 (D) 10

Open solution

Q41. JEE Main 2024 (Online) 27th January Morning Shift

${}^{n-1}C_r = \left(k^2-8\right){}^nC_{r+1}$ if and only if :

(A) $2\sqrt{2} < \mathrm{k} < 2\sqrt{3}$ (B) $2\sqrt{2} < \mathrm{k} \leq 3$

(C) $2\sqrt{3} < \mathrm{k} < 3\sqrt{3}$ (D) $2\sqrt{3} < \mathrm{k} \leq 3\sqrt{2}$

Open solution

Q42. JEE Main 2024 (Online) 27th January Morning Shift

If A denotes the sum of all the coefficients in the expansion of $\left(1-3x+10x^2\right)^{\mathrm{n}}$ and B denotes the sum of all the coefficients in the expansion of $\left(1+x^2\right)^n$, then :

(A) $\mathrm{B} = \mathrm{A}^3$ (B) $3\mathrm{A} = \mathrm{B}$ (C) $A = 3B$ (D) $\mathrm{A} = \mathrm{B}^3$

Open solution

Q43. JEE Main 2023 (Online) 15th April Morning Shift

Let $\left(a+bx+cx^2\right)^{10} = \sum_{i=0}^{20} p_i x^i, a, b, c \in \mathbb{N}$.

If $p_1 = 20$ and $p_2 = 210$, then $2(a+b+c)$ is equal to :

(A) 15 (B) 8 (C) 6 (D) 12

Open solution

Q44. JEE Main 2023 (Online) 13th April Evening Shift

The coefficient of x^5 in the expansion of $\left(2x^3-\frac{1}{3x^2}\right)^5$ is :

(A) $\frac{26}{3}$ (B) $\frac{80}{9}$ (C) 9 (D) 8

Open solution

Q45. JEE Main 2023 (Online) 13th April Morning Shift

Fractional part of the number $\frac{4^{2022}}{15}$ is equal to

(A) $\frac{8}{15}$ (B) $\frac{4}{15}$ (C) $\frac{1}{15}$ (D) $\frac{14}{15}$

Open solution

Q46. JEE Main 2023 (Online) 11th April Evening Shift

The sum of the coefficients of three consecutive terms in the binomial expansion of $(1+\mathrm{x})^{\mathrm{n}+2}$, which are in the ratio $1:3:5$, is equal to :

(A) 63 (B) 92 (C) 25 (D) 41

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q47. JEE Main 2023 (Online) 11th April Evening Shift

If the 1011th term from the end in the binomial expansion of $\left(\frac{4x}{5}-\frac{5}{2x}\right)^{2022}$ is 1024 times 1011th term from the beginning, then $|x|$ is equal to

(A) $\frac{5}{16}$ (B) 8 (C) 12 (D) 15

Open solution

Q48. JEE Main 2023 (Online) 10th April Evening Shift

Let the number $(22)^{2022}+(2022)^{22}$ leave the remainder α when divided by 3 and β when divided by 7. Then $\left(\alpha^2+\beta^2\right)$ is equal to :

(A) 13 (B) 10 (C) 20 (D) 5

Open solution

Q49. JEE Main 2023 (Online) 10th April Evening Shift

If the coefficients of x and x^2 in $(1+x)^{\mathrm{p}}(1-x)^{\mathrm{q}}$ are 4 and -5 respectively, then $2p+3q$ is equal to :

(A) 66 (B) 60 (C) 69 (D) 63

Open solution

Q50. JEE Main 2023 (Online) 10th April Morning Shift

If the coefficient of x^7 in $\left(ax-\frac{1}{bx^2}\right)^{13}$ and the coefficient of x^{-5} in $\left(ax+\frac{1}{bx^2}\right)^{13}$ are equal, then a^4b^4 is equal to :

(A) 22 (B) 33 (C) 44 (D) 11

Open solution

Q51. JEE Main 2023 (Online) 8th April Evening Shift

$25^{190}-19^{190}-8^{190}+2^{190}$ is divisible by :

(A) 14 but not by 34 (B) neither 14 nor 34

(C) both 14 and 34 (D) 34 but not by 14

Open solution

Q52. JEE Main 2023 (Online) 8th April Evening Shift

The absolute difference of the coefficients of x^{10} and x^7 in the expansion of $\left(2x^2+\frac{1}{2x}\right)^{11}$ is equal to :

(A) 11^3-11 (B) 13^3-13 (C) 12^3-12 (D) 10^3-10

Open solution

Q53. JEE Main 2023 (Online) 6th April Evening Shift

If the coefficient of x^7 in $\left(ax^2+\frac{1}{2bx}\right)^{11}$ and x^{-7} in $\left(ax-\frac{1}{3bx^2}\right)^{11}$ are equal, then :

(A) $243ab=64$ (B) $32ab=729$ (C) $64ab=243$ (D) $729ab=32$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q54. JEE Main 2023 (Online) 6th April Evening Shift

Among the statements :

(S1) : $2023^{2022} - 1999^{2022}$ is divisible by 8

(S2) : $13(13)^n - 12n - 13$ is divisible by 144 for infinitely many $n \in \mathbb{N}$

(A) both (S1) and (S2) are incorrect

(B) only (S1) is correct

(C) only (S2) is correct

(D) both (S1) and (S2) are correct

Open solution

Q55. JEE Main 2023 (Online) 6th April Morning Shift

If the ratio of the fifth term from the beginning to the fifth term from the end in the expansion of $\left(\sqrt[4]{2} + \frac{1}{\sqrt[4]{3}}\right)^n$ is $\sqrt{6} : 1$, then the third term from the beginning is :

(A) $30\sqrt{2}$

(B) $60\sqrt{3}$

(C) $60\sqrt{2}$

(D) $30\sqrt{3}$

Open solution

Q56. JEE Main 2023 (Online) 30th January Evening Shift

Let $x = (8\sqrt{3} + 13)^{13}$ and $y = (7\sqrt{2} + 9)^9$. If $[t]$ denotes the greatest integer $\leq t$, then :

(A) $[x]$ is odd but $[y]$ is even

(B) $[x]$ and $[y]$ are both odd

(C) $[x] + [y]$ is even

(D) $[x]$ is even but $[y]$ is odd

Open solution

Q57. JEE Main 2023 (Online) 30th January Morning Shift

If the coefficient of x^{15} in the expansion of $\left(\mathrm{a}x^3 + \frac{1}{\mathrm{b}x^{1/3}}\right)^{15}$ is equal to the coefficient of x^{-15} in the expansion of $\left(ax^{1/3} - \frac{1}{bx^3}\right)^{15}$, where a and b are positive real numbers, then for each such ordered pair (a, b) :

(A) a = 3b

(B) ab = 1

(C) ab = 3

(D) a = b

Open solution

Q58. JEE Main 2023 (Online) 29th January Evening Shift

Let K be the sum of the coefficients of the odd powers of x in the expansion of $(1 + x)^{99}$. Let a be the middle term in the expansion of $\left(2 + \frac{1}{\sqrt{2}}\right)^{200}$. If $\frac{{}^{200}C_{99}K}{a} = \frac{2^l m}{n}$, where m and n are odd numbers, then the ordered pair (l, n) is equal to

(A) (50, 101)

(B) (50, 51)

(C) (51, 101)

(D) (51, 99)

Open solution

Q59. JEE Main 2023 (Online) 25th January Evening Shift

$\sum_{k=0}^{6} {}^{51-k}C_3$ is equal to :

(A) ${}^{51}\mathrm{C}_4 - {}^{45}\mathrm{C}_4$

(B) ${}^{51}\mathrm{C}_3 - {}^{45}\mathrm{C}_3$

(C) ${}^{52}\mathrm{C}_3 - {}^{45}\mathrm{C}_3$

(D) ${}^{52}\mathrm{C}_4 - {}^{45}\mathrm{C}_4$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q60. JEE Main 2022 (Online) 28th July Morning Shift

The remainder when $7^{2022} + 3^{2022}$ is divided by 5 is :

(A) 0 (B) 2 (C) 3 (D) 4

Open solution

Q61. JEE Main 2022 (Online) 27th July Morning Shift

The remainder when $(2021)^{2022} + (2022)^{2021}$ is divided by 7 is

(A) 0 (B) 1 (C) 2 (D) 6

Open solution

Q62. JEE Main 2022 (Online) 25th July Evening Shift

The remainder when $(11)^{1011} + (1011)^{11}$ is divided by 9 is

(A) 1 (B) 4 (C) 6 (D) 8

Open solution

Q63. JEE Main 2022 (Online) 30th June Morning Shift

For two positive real numbers a and b such that $\frac{1}{a^2} + \frac{1}{b^3} = 4$, then minimum value of the constant term in the expansion of $\left(ax^{\frac{1}{8}} + bx^{-\frac{1}{12}}\right)^{10}$ is :

(A) $\frac{105}{2}$ (B) $\frac{105}{4}$ (C) $\frac{105}{8}$ (D) $\frac{105}{16}$

Open solution

Q64. JEE Main 2022 (Online) 29th June Morning Shift

If the constant term in the expansion of

$\left(3x^3 - 2x^2 + \frac{5}{x^5}\right)^{10}$ is $2^k.l$, where l is an odd integer, then the value of k is equal to:

(A) 6 (B) 7 (C) 8 (D) 9

Open solution

Q65. JEE Main 2022 (Online) 28th June Evening Shift

The term independent of x in the expansion of

$(1 - x^2 + 3x^3)\left(\frac{5}{2}x^3 - \frac{1}{5x^2}\right)^{11}$, $x \neq 0$ is :

(A) $\frac{7}{40}$ (B) $\frac{33}{200}$ (C) $\frac{39}{200}$ (D) $\frac{11}{50}$

Open solution

Q66. JEE Main 2022 (Online) 26th June Morning Shift

The remainder when $(2021)^{2023}$ is divided by 7 is :

(A) 1 (B) 2 (C) 5 (D) 6

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q67. JEE Main 2022 (Online) 25th June Evening Shift

The coefficient of x^{101} in the expression $(5+x)^{500}+x(5+x)^{499}+x^2(5+x)^{498}+\ldots\ldots+x^{500}$, x > 0, is

(A) $^{501}C_{101}\,(5)^{399}$ (B) $^{501}C_{101}\,(5)^{400}$ (C) $^{501}C_{100}\,(5)^{400}$ (D) $^{500}C_{101}\,(5)^{399}$

Open solution

Q68. JEE Main 2022 (Online) 24th June Morning Shift

The remainder when 3^{2022} is divided by 5 is :

(A) 1 (B) 2 (C) 3 (D) 4

Open solution

Q69. JEE Main 2021 (Online) 27th July Evening Shift

A possible value of 'x', for which the ninth term in the expansion of $\left\{3^{\log_3\sqrt{25^{x-1}+7}}+3^{\left(-\frac{1}{8}\right)\log_3(5^{x-1}+1)}\right\}^{10}$ in the increasing powers of $3^{\left(-\frac{1}{8}\right)\log_3(5^{x-1}+1)}$ is equal to 180, is :

(A) 0 (B) -1 (C) 2 (D) 1

Open solution

Q70. JEE Main 2021 (Online) 27th July Morning Shift

If the coefficients of x^7 in $\left(x^2+\frac{1}{bx}\right)^{11}$ and x^{-7} in $\left(x-\frac{1}{bx^2}\right)^{11}$, $b\neq 0$, are equal, then the value of b is equal to :

(A) 2 (B) -1 (C) 1 (D) -2

Open solution

Q71. JEE Main 2021 (Online) 25th July Evening Shift

The sum of all those terms which are rational numbers in the expansion of $(2^{1/3}+3^{1/4})^{12}$ is :

(A) 89 (B) 27 (C) 35 (D) 43

Open solution

Q72. JEE Main 2021 (Online) 25th July Evening Shift

If the greatest value of the term independent of 'x' in the expansion of $\left(x\sin\alpha+a\frac{\cos\alpha}{x}\right)^{10}$ is $\frac{10!}{(5!)^2}$, then the value of 'a' is equal to :

(A) -1 (B) 1 (C) -2 (D) 2

Open solution

Q73. JEE Main 2021 (Online) 25th July Evening Shift

The lowest integer which is greater than $\left(1+\frac{1}{10^{100}}\right)^{10^{100}}$ is _____.

(A) 3 (B) 4 (C) 2 (D) 1

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q74. JEE Main 2021 (Online) 25th July Morning Shift

If b is very small as compared to the value of a, so that the cube and other higher powers of $\frac{b}{a}$ can be neglected in the identity $\frac{1}{a-b}+\frac{1}{a-2b}+\frac{1}{a-3b}+\ldots\ldots+\frac{1}{a-nb}=\alpha n+\beta n^2+\gamma n^3$, then the value of γ is :

(A) $\frac{a^2+b}{3a^3}$ (B) $\frac{a+b}{3a^2}$ (C) $\frac{b^2}{3a^3}$ (D) $\frac{a+b^2}{3a^3}$

Open solution

Q75. JEE Main 2021 (Online) 17th March Morning Shift

If the fourth term in the expansion of $\left(x+x^{\log_2 x}\right)^7$ is 4480, then the value of x where x∈N is equal to :

(A) 3 (B) 1 (C) 4 (D) 2

Open solution

Q76. JEE Main 2021 (Online) 16th March Morning Shift

If n is the number of irrational terms in the expansion of $\left(3^{1/4}+5^{1/8}\right)^{60}$, then (n − 1) is divisible by :

(A) 30 (B) 8 (C) 7 (D) 26

Open solution

Q77. JEE Main 2021 (Online) 26th February Morning Shift

The maximum value of the term independent of 't' in the expansion of $\left(tx^{\frac{1}{5}}+\frac{(1-x)^{\frac{1}{10}}}{t}\right)^{10}$ where x∈(0, 1) is :

(A) $\frac{10!}{\sqrt{3}(5!)^2}$ (B) $\frac{2.10!}{3\sqrt{3}(5!)^2}$

(C) $\frac{10!}{3(5!)^2}$ (D) $\frac{2.10!}{3(5!)^2}$

Open solution

Q78. JEE Main 2020 (Online) 6th September Evening Slot

If the constant term in the binomial expansion of $\left(\sqrt{x}-\frac{k}{x^2}\right)^{10}$ is 405, then |k| equals :

(A) 3 (B) 9 (C) 1 (D) 2

Open solution

Q79. JEE Main 2020 (Online) 6th September Morning Slot

If {p} denotes the fractional part of the number p, then $\left\{\frac{3^{200}}{8}\right\}$, is equal to :

(A) $\frac{5}{8}$ (B) $\frac{7}{8}$ (C) $\frac{1}{8}$ (D) $\frac{3}{8}$

Open solution

Q80. JEE Main 2020 (Online) 3rd September Evening Slot

If the term independent of x in the expansion of $\left(\frac{3}{2}x^2-\frac{1}{3x}\right)^9$ is k, then 18 k is equal to :

(A) 5 (B) 9 (C) 7 (D) 11

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q81. JEE Main 2020 (Online) 3rd September Morning Slot

If the number of integral terms in the expansion of $(3^{1/2} + 5^{1/8})^n$ is exactly 33, then the least value of n is :

(A) 264 (B) 256 (C) 128 (D) 248

Open solution

Q82. JEE Main 2020 (Online) 2nd September Morning Slot

Let $\alpha > 0$, $\beta > 0$ be such that $\alpha^3 + \beta^2 = 4$. If the maximum value of the term independent of x in the binomial expansion of $\left(\alpha x^{\frac{1}{9}} + \beta x^{-\frac{1}{6}}\right)^{10}$ is 10k, then k is equal to :

(A) 176 (B) 336 (C) 352 (D) 84

Open solution

Q83. JEE Main 2020 (Online) 9th January Evening Slot

In the expansion of $\left(\frac{x}{\cos\theta} + \frac{1}{x\sin\theta}\right)^{16}$, if ℓ_1 is the least value of the term independent of x when $\frac{\pi}{8} \le \theta \le \frac{\pi}{4}$ and ℓ_2 is the least value of the term independent of x when $\frac{\pi}{16} \le \theta \le \frac{\pi}{8}$, then the ratio $\ell_2 : \ell_1$ is equal to :

(A) 8 : 1 (B) 16 : 1 (C) 1 : 8 (D) 1 : 16

Open solution

Q84. JEE Main 2020 (Online) 8th January Evening Slot

If α and β be the coefficients of x^4 and x^2 respectively in the expansion of $\left(x + \sqrt{x^2-1}\right)^6 + \left(x - \sqrt{x^2-1}\right)^6$, then

(A) $\alpha + \beta = 60$ (B) $\alpha - \beta = 60$ (C) $\alpha + \beta = -30$ (D) $\alpha - \beta = -132$

Open solution

Q85. JEE Main 2020 (Online) 7th January Evening Slot

The coefficient of x^7 in the expression $(1 + x)^{10} + x(1 + x)^9 + x^2(1 + x)^8 + ...+ x^{10}$ is:

(A) 120 (B) 330 (C) 420 (D) 210

Open solution

Q86. JEE Main 2020 (Online) 7th January Morning Slot

The greatest positive integer k, for which $49^k + 1$ is a factor of the sum $49^{125} + 49^{124} + ... + 49^2 + 49 + 1$, is:

(A) 32 (B) 60 (C) 63 (D) 65

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q87. JEE Main 2019 (Online) 12th April Evening Slot

The term independent of x in the expansion of $\left(\frac{1}{60}-\frac{x^8}{81}\right).\left(2x^2-\frac{3}{x^2}\right)^6$ is equal to :

(A) 36 (B) - 108 (C) - 36 (D) - 72

Open solution

Q88. JEE Main 2019 (Online) 12th April Morning Slot

The coefficient of x^{18} in the product $(1 + x)(1 - x)^{10}(1 + x + x^2)^9$ is :

(A) 126 (B) - 84 (C) - 126 (D) 84

Open solution

Q89. JEE Main 2019 (Online) 10th April Evening Slot

The smallest natural number n, such that the coefficient of x in the expansion of $\left(x^2+\frac{1}{x^3}\right)^n$ is ${}^nC_{23}$, is :

(A) 23 (B) 58 (C) 38 (D) 35

Open solution

Q90. JEE Main 2019 (Online) 9th April Morning Slot

If the fourth term in the binomial expansion of $\left(\frac{2}{x}+x^{\log_8 x}\right)^6$ $(x > 0)$ is 20×8^7, then a value of x is :

(A) 8^{-2} (B) 8^2 (C) 8^3 (D) 8

Open solution

Q91. JEE Main 2019 (Online) 8th April Evening Slot

If the fourth term in the binomial expansion of $\left(\sqrt{x^{\left(\frac{1}{1+\log_{10}x}\right)}}+x^{\frac{1}{12}}\right)^6$ is equal to 200, and $x > 1$, then the value of x is :

(A) 100 (B) 10^3 (C) 10 (D) 10^4

Open solution

Q92. JEE Main 2019 (Online) 12th January Evening Slot

The total number of irrational terms in the binomial expansion of $(7^{1/5} - 3^{1/10})^{60}$ is :

(A) 54 (B) 55 (C) 49 (D) 48

Open solution

Q93. JEE Main 2019 (Online) 12th January Morning Slot

A ratio of the 5th term from the beginning to the 5th term from the end in the binomial expansion of $\left(2^{1/3}+\frac{1}{2(3)^{1/3}}\right)^{10}$ is :

(A) $1 : 2(6)^{1/3}$ (B) $1 : 4(6)^{1/3}$ (C) $2(36)^{1/3} : 1$ (D) $4(36)^{1/3} : 1$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q94. JEE Main 2019 (Online) 11th January Morning Slot

The sum of the real values of x for which the middle term in the binomial expansion of $\left(\frac{x^3}{3}+\frac{3}{x}\right)^8$ equals 5670 is :

(A) 0 (B) 8 (C) 6 (D) 4

Open solution

Q95. JEE Main 2019 (Online) 10th January Evening Slot

If $\sum_{r=0}^{25}\left\{{}^{50}C_r.\ {}^{50-r}C_{25-r}\right\}=K\left({}^{50}C_{25}\right)$, then K is equal to :

(A) 2^{24} (B) $2^{25}-1$ (C) 2^{25} (D) $(25)^2$

Open solution

Q96. JEE Main 2019 (Online) 10th January Evening Slot

The positive value of λ for which the co-efficient of x^2 in the expression $x^2\left(\sqrt{x}+\frac{\lambda}{x^2}\right)^{10}$ is 720, is -

(A) 4 (B) $2\sqrt{2}$ (C) 3 (D) $\sqrt{5}$

Open solution

Q97. JEE Main 2019 (Online) 10th January Morning Slot

If the third term in the binomial expansion of $\left(1+x^{\log_2 x}\right)^5$ equals 2560, then a possible value of x is -

(A) $2\sqrt{2}$ (B) $4\sqrt{2}$ (C) $\frac{1}{8}$ (D) $\frac{1}{4}$

Open solution

Q98. JEE Main 2019 (Online) 9th January Morning Slot

If the fractional part of the number $\left\{\frac{2^{403}}{15}\right\}$ is $\frac{k}{15}$, then k is equal to :

(A) 8 (B) 14 (C) 6 (D) 1

Open solution

Q99. JEE Main 2017 (Online) 9th April Morning Slot

The coefficient of x^{-5} in the binomial expansion of

$\left(\frac{x+1}{x^{\frac{2}{3}}-x^{\frac{1}{3}}+1}-\frac{x-1}{x-x^{\frac{1}{2}}}\right)^{10}$, where $x\neq 0, 1$, is :

(A) 1 (B) 4 (C) -4 (D) -1

Open solution

Q100. JEE Main 2017 (Online) 8th April Morning Slot

If $(27)^{999}$ is divided by 7, then the remainder is :

(A) 1 (B) 2 (C) 3 (D) 6

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q101. JEE Main 2016 (Online) 10th April Morning Slot

If the coefficients of x^{-2} and x^{-4} in the expansion of $\left(x^{\frac{1}{3}}+\frac{1}{2x^{\frac{1}{3}}}\right)^{18}, (x>0)$, are m and n respectively, then $\frac{m}{n}$ is equal to :

(A) 182 (B) $\frac{4}{5}$ (C) $\frac{5}{4}$ (D) 27

Open solution

Q102. JEE Main 2016 (Online) 9th April Morning Slot

For $x \in \mathbf{R}$, $x \neq -1$,

if $(1+x)^{2016} + x(1+x)^{2015} + x^2(1+x)^{2014} + \ldots + x^{2016} = \sum_{i=0}^{2016} a_i x^i$, then a_{17} is equal to :

(A) $\frac{2017!}{17!\ 2000!}$ (B) $\frac{2016!}{17!\ 1999!}$ (C) $\frac{2017!}{2000!}$ (D) $\frac{2016!}{16!}$

Open solution

Q103. JEE Main 2016 (Online) 9th April Morning Slot

The value of $\sum_{r=1}^{15} r^2 \left(\frac{^{15}C_r}{^{15}C_{r-1}}\right)$ is equal to :

(A) 560 (B) 680 (C) 1240 (D) 1085

Open solution

Q104. JEE Main 2015 (Online) 11th April Morning Slot

The term independent of x in the binomial expansion of

$(1-\frac{1}{x}+3x^5)\left(2x^2-\frac{1}{x}\right)^8$

(A) 400 (B) 496 (C) -400 (D) -496

Open solution

Q105. JEE Main 2014 (Online) 19th April Morning Slot

The coefficient of x^{1012} in the expansion of $(1+x^n+x^{253})^{10}$, (where $n \leq 22$ is any positive integer), is :

(A) 1 (B) $^{10}C_4$ (C) 4n (D) $^{253}C_4$

Open solution

Q106. JEE Main 2014 (Online) 12th April Morning Slot

If $\left(2+\frac{x}{3}\right)^{55}$ is expanded in the ascending powers of x and the coefficients of powers of x in two consecutive terms of the expansion are equal, then these terms are :

(A) 7th and 8th (B) 8th and 9th (C) 28th and 29th (D) 27th and 28th

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q107. JEE Main 2014 (Online) 12th April Morning Slot

If $1 + x^4 + x^5 = \sum_{i=0}^{5} a_i (1 + x)^i$, for all x in **R**, then a_2 is :

(A) -4 (B) 6 (C) -8 (D) 10

Open solution

Q108. JEE Main 2014 (Online) 9th April Morning Slot

The number of terms in the expansion of

$(1 + x)^{101} (1 + x^2 - x)^{100}$ in powers of x is :

(A) 302 (B) 301 (C) 202 (D) 101

Open solution

Q109. JEE Main 2013 (Online) 23th April Morning Slot

The sum of the rational terms in the binomial expansion of $\left(2^{\frac{1}{2}} + 3^{\frac{1}{5}}\right)^{10}$ is

(A) 25 (B) 32 (C) 9 (D) 41

Open solution

Q110. JEE Main 2013 (Online) 22th April Morning Slot

If the 7th terms in the binomial expansion of $\left(\frac{3}{\sqrt[3]{84}} + \sqrt{3}\log x\right)^9$, x > 0, is equal to 729, then x can be:

(A) e^2 (B) e (C) $\frac{e}{2}$ (D) 2e

Open solution

Q111. JEE Main 2013 (Online) 9th April Morning Slot

The ratio of the coefficient of x^{15} to the term independent of x in the expansion of $\left(x^2 + \frac{2}{x}\right)^{15}$ is

(A) 7 : 16 (B) 7 : 64 (C) 1 : 4 (D) 1 : 32

Open solution

Q112. JEE Main 2013 (Offline)

The term independent of x in expansion of

$\left(\frac{x+1}{x^{2/3}-x^{1/3}+1} - \frac{x-1}{x-x^{1/2}}\right)^{10}$ is

(A) 4 (B) 120 (C) 210 (D) 310

Open solution

Q113. AIEEE 2012 (Online) 26th May

The middle term in the expansion of $\left(1 - \frac{1}{x}\right)^n (1 - x)^n$ in powers of x is

(A) $-{}^{2n}C_{n-1}$ (B) $-{}^{2n}C_n$ (C) ${}^{2n}C_{n-1}$ (D) ${}^{2n}C_n$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q114. AIEEE 2012 (Online) 12th May

The number of terms in the expansion of $\left(y^{1/5}+x^{1/10}\right)^{55}$, in which powers of x and y are free from radical signs are

(A) six (B) twelve (C) seven (D) five

Open solution

Q115. AIEEE 2012

If n is a positive integer, then $\left(\sqrt{3}+1\right)^{2n}-\left(\sqrt{3}-1\right)^{2n}$ is :

(A) an irrational number

(B) an odd positive integer

(C) an even positive integer

(D) a rational number other than positive integers

Open solution

Q116. AIEEE 2009

The remainder left out when $8^{2n}-(62)^{2n+1}$ is divided by 9 is :

(A) 2 (B) 7 (C) 8 (D) 0

Open solution

Q117. AIEEE 2007

In the binomial expansion of $(a-b)^n,\ \ n\geq 5$, the sum of 5th and 6th terms is zero, then a/b equals

(A) $\frac{n-5}{6}$ (B) $\frac{n-4}{5}$ (C) $\frac{5}{n-4}$ (D) $\frac{6}{n-5}$

Open solution

Q118. AIEEE 2006

If the expansion in powers of x of the function $\frac{1}{(1-ax)(1-bx)}$ is $a_0+a_1x+a_2x^2+a_3x^3.....$ then a_n is

(A) $\frac{b^n-a^n}{b-a}$ (B) $\frac{a^n-b^n}{b-a}$

(C) $\frac{a^{n+1}-b^{n+1}}{b-a}$ (D) $\frac{b^{n+1}-a^{n+1}}{b-a}$

Open solution

Q119. AIEEE 2005

If the coefficient of x^7 in $\left[ax^2+\left(\frac{1}{bx}\right)\right]^{11}$ equals the coefficient of x^{-7} in $\left[ax-\left(\frac{1}{bx^2}\right)\right]^{11}$, then a and b satisfy the relation

(A) $a-b=1$ (B) $a+b=1$ (C) $\frac{a}{b}=1$ (D) $ab=1$

Open solution

Q120. AIEEE 2005

If x is so small that x^3 and higher powers of x may be neglected, then $\frac{(1+x)^{\frac{3}{2}}-\left(1+\frac{1}{2}x\right)^3}{(1-x)^{\frac{1}{2}}}$ may be approximated as

(A) $1-\frac{3}{8}x^2$ (B) $3x+\frac{3}{8}x^2$ (C) $-\frac{3}{8}x^2$ (D) $\frac{x}{2}-\frac{3}{8}x^2$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q121. AIEEE 2005

If the coefficients of rth, $(r+1)^{th}$, and $(r+2)^{th}$ terms in the binomial expansion of $(1+y)^m$ are in A.P., then m and r satisfy the equation

(A) $m^2 - m(4r-1) + 4r^2 - 2 = 0$
(B) $m^2 - m(4r+1) + 4r^2 + 2 = 0$
(C) $m^2 - m(4r+1) + 4r^2 - 2 = 0$
(D) $m^2 - m(4r-1) + 4r^2 + 2 = 0$

Open solution

Q122. AIEEE 2004

The coefficient of the middle term in the binomial expansion in powers of x of $(1+\alpha x)^4$ and $(1-\alpha x)^6$ is the same if α equals

(A) $\frac{3}{5}$
(B) $\frac{10}{3}$
(C) $\frac{-3}{10}$
(D) $\frac{-5}{3}$

Open solution

Q123. AIEEE 2004

The coefficient of x^n in expansion of $(1+x)(1-x)^n$ is

(A) $(-1)^{n-1}n$
(B) $(-1)^n(1-n)$
(C) $(-1)^{n-1}(n-1)^2$
(D) $(n-1)$

Open solution

Q124. AIEEE 2003

If x is positive, the first negative term in the expansion of $(1+x)^{27/5}$ is

(A) 6th term
(B) 7th term
(C) 5th term
(D) 8th term.

Open solution

Q125. AIEEE 2003

The number of integral terms in the expansion of $\left(\sqrt{3}+\sqrt[8]{5}\right)^{256}$ is

(A) 35
(B) 32
(C) 33
(D) 34

Open solution

Q126. AIEEE 2002

The coefficients of x^p and x^q in the expansion of $(1+x)^{p+q}$ are

(A) equal
(B) equal with opposite signs
(C) reciprocals of each other
(D) none of these

Open solution

Q127. AIEEE 2002

The positive integer just greater than $(1+0.0001)^{10000}$ is

(A) 4
(B) 5
(C) 2
(D) 3

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Numerical 69 questions

Q128. JEE Main 2026 (Online) 6th April Evening Shift

If $\left(1-x^3\right)^{10}=\sum_{r=0}^{10} a_r x^r(1-x)^{30-2r}$, then $\frac{9a_9}{a_{10}}$ is equal to ____ .

Open solution

Q129. JEE Main 2026 (Online) 5th April Morning Shift

If the sum of the coefficients of x^7 and x^{14} in the expansion of $\left(\frac{1}{x^3}-x^4\right)^n, x \neq 0$, is zero, then the value of n is ____ .

Open solution

Q130. JEE Main 2026 (Online) 21st January Evening Shift

If $\left(\frac{1}{{}^{15}C_0}+\frac{1}{{}^{15}C_1}\right)\left(\frac{1}{{}^{15}C_1}+\frac{1}{{}^{15}C_2}\right)\ldots\left(\frac{1}{{}^{15}C_{12}}+\frac{1}{{}^{15}C_{13}}\right)=\frac{\alpha^{13}}{{}^{14}C_0\,{}^{14}C_1\ldots{}^{14}C_{12}}$, then 30α is equal to ____ .

Open solution

Q131. JEE Main 2025 (Online) 8th April Evening Shift

The product of the last two digits of $(1919)^{1919}$ is

Open solution

Q132. JEE Main 2025 (Online) 7th April Evening Shift

The sum of the series
$2\times1\times{}^{20}C_4-3\times2\times{}^{20}C_5+4\times3\times{}^{20}C_6-5\times4\times{}^{20}C_7+\cdots\cdots+18\times17\times{}^{20}C_{20}$, is equal to _____.

Open solution

Q133. JEE Main 2025 (Online) 3rd April Evening Shift

Let $\left(1+x+x^2\right)^{10}=a_0+a_1x+a_2x^2+\ldots+a_{20}x^{20}$. If $(a_1+a_3+a_5+\ldots+a_{19})-11a_2=121k$, then k is equal to_____ .

Open solution

Q134. JEE Main 2025 (Online) 28th January Morning Shift

If $\alpha=1+\sum_{r=1}^{6}(-3)^{r-1}{}^{12}C_{2r-1}$, then the distance of the point $(12,\sqrt{3})$ from the line $\alpha x-\sqrt{3}y+1=0$ is _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q135. JEE Main 2025 (Online) 23rd January Morning Shift

The sum of all rational terms in the expansion of $\left(1+2^{1/3}+3^{1/2}\right)^6$ is equal to _____.

Open solution

Q136. JEE Main 2025 (Online) 22nd January Evening Shift

If $\sum_{r=1}^{30} \frac{r^2\left({}^{30}C_r\right)^2}{{}^{30}C_{r-1}} = \alpha \times 2^{29}$, then α is equal to _____.

Open solution

Q137. JEE Main 2025 (Online) 22nd January Morning Shift

If $\sum_{r=0}^{5} \frac{{}^{11}C_{2r+1}}{2r+2} = \frac{\mathrm{m}}{\mathrm{n}}, \gcd(\mathrm{m},\mathrm{n}) = 1$, then $\mathrm{m}-\mathrm{n}$ is equal to _____.

Open solution

Q138. JEE Main 2024 (Online) 9th April Morning Shift

The remainder when 428^{2024} is divided by 21 is _____.

Open solution

Q139. JEE Main 2024 (Online) 8th April Morning Shift

Let $\alpha = \sum_{r=0}^{n} \left(4r^2+2r+1\right){}^nC_r$ and $\beta = \left(\sum_{r=0}^{n} \frac{{}^nC_r}{r+1}\right) + \frac{1}{n+1}$. If $140 < \frac{2\alpha}{\beta} < 281$, then the value of n is _____.

Open solution

Q140. JEE Main 2024 (Online) 6th April Morning Shift

If the second, third and fourth terms in the expansion of $(x+y)^n$ are 135, 30 and $\frac{10}{3}$, respectively, then $6\left(n^3+x^2+y\right)$ is equal to _____.

Open solution

Q141. JEE Main 2024 (Online) 5th April Morning Shift

If the constant term in the expansion of $\left(1+2x-3x^3\right)\left(\frac{3}{2}x^2-\frac{1}{3x}\right)^9$ is p, then $108\mathrm{p}$ is equal to _____.

Open solution

Q142. JEE Main 2024 (Online) 4th April Morning Shift

Let $a = 1+\frac{{}^2\mathrm{C}_2}{3!}+\frac{{}^3\mathrm{C}_2}{4!}+\frac{{}^4\mathrm{C}_2}{5!}+\ldots, \mathrm{b} = 1+\frac{{}^1\mathrm{C}_0+{}^1\mathrm{C}_1}{1!}+\frac{{}^2\mathrm{C}_0+{}^2\mathrm{C}_1+{}^2\mathrm{C}_2}{2!}+\frac{{}^3\mathrm{C}_0+{}^3\mathrm{C}_1+{}^3\mathrm{C}_2+{}^3\mathrm{C}_3}{3!}+\ldots$. Then $\frac{2b}{a^2}$ is equal to _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q143. JEE Main 2024 (Online) 1st February Morning Shift

If the coefficient of x^{30} in the expansion of $\left(1+\frac{1}{x}\right)^6\left(1+x^2\right)^7\left(1-x^3\right)^8; x \neq 0$ is α, then $|\alpha|$ equals _____.

Open solution

Q144. JEE Main 2024 (Online) 31st January Evening Shift

Let the coefficient of x^r in the expansion of $(x+3)^{n-1}+(x+3)^{n-2}(x+2)+(x+3)^{n-3}(x+2)^2+\ldots\ldots\ldots+(x+2)^{n-1}$ be α_r. If $\sum_{r=0}^{n}\alpha_r=\beta^n-\gamma^n, \beta, \gamma \in \mathbb{N}$, then the value of $\beta^2+\gamma^2$ equals _____.

Open solution

Q145. JEE Main 2024 (Online) 31st January Morning Shift

In the expansion of $(1+x)\left(1-x^2\right)\left(1+\frac{3}{x}+\frac{3}{x^2}+\frac{1}{x^3}\right)^5, x \neq 0$, the sum of the coefficients of x^3 and x^{-13} is equal to _____.

Open solution

Q146. JEE Main 2024 (Online) 30th January Evening Shift

Let $\alpha=\sum_{k=0}^{n}\left(\frac{\left({}^nC_k\right)^2}{k+1}\right)$ and $\beta=\sum_{k=0}^{n-1}\left(\frac{{}^nC_k\,{}^nC_{k+1}}{k+2}\right)$ If $5\alpha=6\beta$, then n equals _____.

Open solution

Q147. JEE Main 2024 (Online) 30th January Morning Shift

Number of integral terms in the expansion of $\left\{7^{\left(\frac{1}{2}\right)}+11^{\left(\frac{1}{6}\right)}\right\}^{824}$ is equal to _____.

Open solution

Q148. JEE Main 2024 (Online) 29th January Evening Shift

Remainder when $64^{32^{32}}$ is divided by 9 is equal to _____.

Open solution

Q149. JEE Main 2024 (Online) 29th January Morning Shift

If $\frac{{}^{11}C_1}{2}+\frac{{}^{11}C_2}{3}+\ldots+\frac{{}^{11}C_9}{10}=\frac{n}{m}$ with $\gcd(n, m)=1$, then $n+m$ is equal to _____.

Open solution

Q150. JEE Main 2024 (Online) 27th January Evening Shift

The coefficient of x^{2012} in the expansion of $(1-x)^{2008}\left(1+x+x^2\right)^{2007}$ is equal to _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q151. JEE Main 2023 (Online) 13th April Evening Shift

The remainder, when 7^{103} is divided by 17, is _____

Open solution

Q152. JEE Main 2023 (Online) 13th April Morning Shift

Let α be the constant term in the binomial expansion of $\left(\sqrt{x}-\frac{6}{x^{\frac{3}{2}}}\right)^{n}, n \leq 15$. If the sum of the coefficients of the remaining terms in the expansion is 649 and the coefficient of x^{-n} is $\lambda\alpha$, then λ is equal to _____.

Open solution

Q153. JEE Main 2023 (Online) 11th April Morning Shift

The mean of the coefficients of $x, x^2, \ldots, x^7$ in the binomial expansion of $(2+x)^9$ is _____.

Open solution

Q154. JEE Main 2023 (Online) 11th April Morning Shift

The number of integral terms in the expansion of $\left(3^{\frac{1}{2}}+5^{\frac{1}{4}}\right)^{680}$ is equal to _____.

Open solution

Q155. JEE Main 2023 (Online) 10th April Morning Shift

The coefficient of x^7 in $(1-x+2x^3)^{10}$ is _____.

Open solution

Q156. JEE Main 2023 (Online) 8th April Morning Shift

Let $[t]$ denote the greatest integer $\leq t$. If the constant term in the expansion of $\left(3x^2-\frac{1}{2x^5}\right)^7$ is α, then $[\alpha]$ is equal to _____.

Open solution

Q157. JEE Main 2023 (Online) 8th April Morning Shift

The largest natural number n such that 3^n divides $66!$ is _____.

Open solution

Q158. JEE Main 2023 (Online) 1st February Evening Shift

Let the sixth term in the binomial expansion of $\left(\sqrt{2^{\log_2(10-3^x)}}+\sqrt[5]{2^{(x-2)\log_2 3}}\right)^{m}$ in the increasing powers of $2^{(x-2)\log_2 3}$, be 21 . If the binomial coefficients of the second, third and fourth terms in the expansion are respectively the first, third and fifth terms of an A.P., then the sum of the squares of all possible values of x is _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q159. JEE Main 2023 (Online) 1st February Evening Shift

If the term without x in the expansion of $\left(x^{\frac{2}{3}}+\frac{\alpha}{x^3}\right)^{22}$ is 7315 , then $|\alpha|$ is equal to _____.

Open solution

Q160. JEE Main 2023 (Online) 1st February Morning Shift

The remainder, when $19^{200}+23^{200}$ is divided by 49 , is _____.

Open solution

Q161. JEE Main 2023 (Online) 31st January Evening Shift

The coefficient of x^{-6}, in the

expansion of $\left(\frac{4x}{5}+\frac{5}{2x^2}\right)^9$, is

Open solution

Q162. JEE Main 2023 (Online) 31st January Evening Shift

If the constant term in the binomial expansion of $\left(\frac{x^{\frac{5}{2}}}{2}-\frac{4}{x^l}\right)^9$ is -84 and the coefficient of x^{-3l} is $2^\alpha\beta$, where $\beta<0$ is an odd number, then $|\alpha l-\beta|$ is equal to _____.

Open solution

Q163. JEE Main 2023 (Online) 31st January Morning Shift

The remainder on dividing 5^{99} by 11 is _____.

Open solution

Q164. JEE Main 2023 (Online) 31st January Morning Shift

Let $\alpha>0$, be the smallest number such that the expansion of $\left(x^{\frac{2}{3}}+\frac{2}{x^3}\right)^{30}$ has a term $\beta x^{-\alpha}, \beta\in\mathbb{N}$. Then α is equal to _____.

Open solution

Q165. JEE Main 2023 (Online) 30th January Evening Shift

50th root of a number x is 12 and 50th root of another number y is 18 . Then the remainder obtained on dividing $(x+y)$ by 25 is _____.

Open solution

Q166. JEE Main 2023 (Online) 29th January Morning Shift

Let the coefficients of three consecutive terms in the binomial expansion of $(1+2x)^n$ be in the ratio 2 : 5 : 8. Then the coefficient of the term, which is in the middle of those three terms, is _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q167. JEE Main 2023 (Online) 29th January Morning Shift

If the co-efficient of x^9 in $\left(\alpha x^3+\frac{1}{\beta x}\right)^{11}$ and the co-efficient of x^{-9} in $\left(\alpha x-\frac{1}{\beta x^3}\right)^{11}$ are equal, then $(\alpha\beta)^2$ is equal to _____.

Open solution

Q168. JEE Main 2023 (Online) 25th January Evening Shift

The remainder when $(2023)^{2023}$ is divided by 35 is _____.

Open solution

Q169. JEE Main 2023 (Online) 25th January Morning Shift

The constant term in the expansion of $\left(2x+\frac{1}{x^7}+3x^2\right)^5$ is _____.

Open solution

Q170. JEE Main 2023 (Online) 24th January Evening Shift

Let the sum of the coefficients of the first three terms in the expansion of $\left(x-\frac{3}{x^2}\right)^n, x\neq 0. n\in\mathbb{N}$, be 376. Then the coefficient of x^4 is _____.

Open solution

Q171. JEE Main 2022 (Online) 30th July Evening Shift

If p_1 and p_2, respectively, are the second largest and the fourth largest prime numbers in the factorization of the coefficient of x^{30} in the expansion of $\left(x^2+2+\frac{1}{x^2}\right)^{-5}\left(1+x^2\right)^{40}$, then p_1+p_2 is equal to _____

Open solution

Q172. JEE Main 2022 (Online) 29th July Morning Shift

Let the ratio of the fifth term from the beginning to the fifth term from the end in the binomial expansion of $\left(\sqrt[4]{2}+\frac{1}{\sqrt[4]{3}}\right)^n$, in the increasing powers of $\frac{1}{\sqrt[4]{3}}$ be $\sqrt[4]{6}:1$. If the sixth term from the beginning is $\frac{\alpha}{\sqrt[4]{3}}$, then α is equal to _____.

Open solution

Q173. JEE Main 2022 (Online) 28th July Evening Shift

Let the coefficients of the middle terms in the expansion of $\left(\frac{1}{\sqrt{6}}+\beta x\right)^4$, $(1-3\beta x)^2$ and $\left(1-\frac{\beta}{2}x\right)^6, \beta>0$, respectively form the first three terms of an A.P. If d is the common difference of this A.P. , then $50-\frac{2d}{\beta^2}$ is equal to _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q174. JEE Main 2022 (Online) 27th July Evening Shift

Let for the 9th term in the binomial expansion of $(3+6x)^n$, in the increasing powers of $6x$, to be the greatest for $x=\frac{3}{2}$, the least value of n is n_0. If k is the ratio of the coefficient of x^6 to the coefficient of x^3, then $k+n_0$ is equal to :

Open solution

Q175. JEE Main 2022 (Online) 25th July Morning Shift

If the maximum value of the term independent of t in the expansion of $\left(t^2x^{\frac{1}{5}}+\frac{(1-x)^{\frac{1}{10}}}{t}\right)^{15}, x\geqslant 0$, is K, then $8K$ is equal to _____.

Open solution

Q176. JEE Main 2022 (Online) 29th June Evening Shift

Let the coefficients of x^{-1} and x^{-3} in the expansion of $\left(2x^{\frac{1}{5}}-\frac{1}{x^{\frac{1}{5}}}\right)^{15}, x>0$, be m and n respectively. If r is a positive integer such that $mn^2={}^{15}C_r\,.\,2^r$, then the value of r is equal to _____.

Open solution

Q177. JEE Main 2022 (Online) 28th June Morning Shift

The number of positive integers k such that the constant term in the binomial expansion of $\left(2x^3+\frac{3}{x^k}\right)^{12}$, $x\neq 0$ is 2^8 . l, where l is an odd integer, is _____.

Open solution

Q178. JEE Main 2022 (Online) 27th June Morning Shift

If the coefficient of x^{10} in the binomial expansion of $\left(\frac{\sqrt{x}}{5^{\frac{1}{4}}}+\frac{\sqrt{5}}{x^{\frac{1}{3}}}\right)^{60}$ is $5^k\,.\,l$, where $l, k\in N$ and l is co-prime to 5, then k is equal to _____.

Open solution

Q179. JEE Main 2022 (Online) 25th June Evening Shift

If the sum of the co-efficient of all the positive even powers of x in the binomial expansion of $\left(2x^3+\frac{3}{x}\right)^{10}$ is $5^{10}-\beta\,.\,3^9$, then β is equal to _____.

Open solution

Q180. JEE Main 2022 (Online) 24th June Evening Shift

The remainder on dividing $1 + 3 + 3^2 + 3^3 + ... + 3^{2021}$ by 50 is _____.

Open solution

Q181. JEE Main 2021 (Online) 31st August Evening Shift

If the coefficient of a^7b^8 in the expansion of $(a + 2b + 4ab)^{10}$ is $K.2^{16}$, then K is equal to _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q182. JEE Main 2021 (Online) 31st August Morning Shift

If $\left(\frac{3^6}{4^4}\right)k$ is the term, independent of x, in the binomial expansion of $\left(\frac{x}{4}-\frac{12}{x^2}\right)^{12}$, then k is equal to _____.

Open solution

Q183. JEE Main 2021 (Online) 27th August Evening Shift

$3 \times 7^{22} + 2 \times 10^{22} - 44$ when divided by 18 leaves the remainder _____.

Open solution

Q184. JEE Main 2021 (Online) 25th July Evening Shift

If the co-efficient of x^7 and x^8 in the expansion of $\left(2+\frac{x}{3}\right)^n$ are equal, then the value of n is equal to _____.

Open solution

Q185. JEE Main 2021 (Online) 25th July Morning Shift

The ratio of the coefficient of the middle term in the expansion of $(1 + x)^{20}$ and the sum of the coefficients of two middle terms in expansion of $(1 + x)^{19}$ is _____.

Open solution

Q186. JEE Main 2021 (Online) 25th July Morning Shift

The term independent of 'x' in the expansion of $\left(\frac{x+1}{x^{2/3}-x^{1/3}+1}-\frac{x-1}{x-x^{1/2}}\right)^{10}$, where $x \neq 0, 1$ is equal to _____.

Open solution

Q187. JEE Main 2021 (Online) 22th July Evening Shift

If the constant term, in binomial expansion of $\left(2x^r+\frac{1}{x^2}\right)^{10}$ is 180, then r is equal to _____.

Open solution

Q188. JEE Main 2021 (Online) 20th July Morning Shift

The number of rational terms in the binomial expansion of $\left(4^{\frac{1}{4}}+5^{\frac{1}{6}}\right)^{120}$ is _____.

Open solution

Q189. JEE Main 2021 (Online) 18th March Evening Shift

The term independent of x in the expansion of

$\left[\frac{x+1}{x^{2/3}-x^{1/3}+1}-\frac{x-1}{x-x^{1/2}}\right]^{10}$, $x \neq 1$, is equal to _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q190. JEE Main 2021 (Online) 17th March Evening Shift

Let the coefficients of third, fourth and fifth terms in the expansion of $\left(x+\frac{a}{x^2}\right)^n, x \neq 0$, be in the ratio 12 : 8 : 3. Then the term independent of x in the expansion, is equal to _____.

Open solution

Q191. JEE Main 2021 (Online) 17th March Morning Shift

If $(2021)^{3762}$ is divided by 17, then the remainder is _____.

Open solution

Q192. JEE Main 2021 (Online) 25th February Evening Shift

If the remainder when x is divided by 4 is 3, then the remainder when $(2020 + x)^{2022}$ is divided by 8 is _____.

Open solution

Q193. JEE Main 2021 (Online) 25th February Evening Shift

The total number of two digit numbers 'n', such that $3^n + 7^n$ is a multiple of 10, is _____.

Open solution

Q194. JEE Main 2020 (Online) 5th September Morning Slot

The natural number m, for which the coefficient of x in the binomial expansion of

$\left(x^m+\frac{1}{x^2}\right)^{22}$ is 1540, is .....

Open solution

Q195. JEE Main 2020 (Online) 4th September Morning Slot

Let $\left(2x^2+3x+4\right)^{10}=\sum_{r=0}^{20} a_r x^r$

Then $\frac{a_7}{a_{13}}$ is equal to _____.

Open solution

Q196. JEE Main 2020 (Online) 9th January Morning Slot

The coefficient of x^4 is the expansion of $(1 + x + x^2)^{10}$ is ____.

Open solution

Out of syllabus 75 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q197. JEE Main 2023 (Online) 12th April Morning Shift · MCQ (Single Correct Answer)

If $\frac{1}{n+1}{}^{n}C_{n}+\frac{1}{n}{}^{n}C_{n-1}+\ldots+\frac{1}{2}{}^{n}C_{1}+{}^{n}C_{0}=\frac{1023}{10}$ then n is equal to :

(A) 9 (B) 6 (C) 7 (D) 8

Open solution

Q198. JEE Main 2023 (Online) 12th April Morning Shift · MCQ (Single Correct Answer)

The sum, of the coefficients of the first 50 terms in the binomial expansion of $(1-x)^{100}$, is equal to

(A) ${}^{99}C_{49}$ (B) ${}^{101}C_{50}$ (C) $-{}^{99}C_{49}$ (D) $-{}^{101}C_{50}$

Open solution

Q199. JEE Main 2023 (Online) 8th April Morning Shift · MCQ (Single Correct Answer)

If the coefficients of three consecutive terms in the expansion of $(1+x)^{n}$ are in the ratio $1:5:20$, then the coefficient of the fourth term is

(A) 3654 (B) 1827 (C) 5481 (D) 2436

Open solution

Q200. JEE Main 2023 (Online) 6th April Morning Shift · MCQ (Single Correct Answer)

If ${}^{2n}C_{3}:{}^{n}C_{3}=10:1$, then the ratio $\left(n^{2}+3n\right):\left(n^{2}-3n+4\right)$ is :

(A) $27:11$ (B) $2:1$ (C) $35:16$ (D) $65:37$

Open solution

Q201. JEE Main 2023 (Online) 30th January Morning Shift · MCQ (Single Correct Answer)

The coefficient of x^{301} in $(1+x)^{500}+x(1+x)^{499}+x^{2}(1+x)^{498}+\ldots+x^{500}$ is :

(A) ${}^{500}C_{300}$ (B) ${}^{501}C_{200}$ (C) ${}^{500}C_{301}$ (D) ${}^{501}C_{302}$

Open solution

Q202. JEE Main 2023 (Online) 25th January Morning Shift · MCQ (Single Correct Answer)

If a_r is the coefficient of x^{10-r} in the Binomial expansion of $(1+x)^{10}$, then $\sum_{r=1}^{10} r^{3}\left(\frac{a_r}{a_{r-1}}\right)^{2}$ is equal to

(A) 3025 (B) 4895 (C) 5445 (D) 1210

Open solution

Q203. JEE Main 2023 (Online) 24th January Evening Shift · MCQ (Single Correct Answer)

If $({}^{30}C_{1})^{2}+2({}^{30}C_{2})^{2}+3({}^{30}C_{3})^{2}+\ldots+30({}^{30}C_{30})^{2}=\frac{\alpha 60!}{(30!)^{2}}$ then α is equal to :

(A) 30 (B) 10 (C) 15 (D) 60

Open solution

Q204. JEE Main 2023 (Online) 24th January Morning Shift · MCQ (Single Correct Answer)

The value of $\sum_{r=0}^{22} {}^{22}C_{r}\,{}^{23}C_{r}$ is

(A) ${}^{44}C_{23}$ (B) ${}^{45}C_{23}$ (C) ${}^{44}C_{22}$ (D) ${}^{45}C_{24}$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q205. JEE Main 2022 (Online) 30th July Evening Shift · MCQ (Single Correct Answer)

The coefficient of x^{1011} in the expression
$(1+2x)^{2022}+2x(1+2x)^{2021}+(2x)^2(1+2x)^{2020}+\ldots+(2x)^{2022}$ is

(A) ${}^{2022}C_{1011}$ (B) ${}^{2022}C_{1011}\times(2)^{1011}$

(C) ${}^{2023}C_{1011}\times(2)^{1011}$ (D) ${}^{2023}C_{1011}$

Open solution

Q206. JEE Main 2022 (Online) 29th July Evening Shift · MCQ (Single Correct Answer)

$\sum_{r=1}^{20}(r^2+1)(r!)$ is equal to

(A) $22!-21!$ (B) $22!-2(21!)$ (C) $21!-2(20!)$ (D) $21!-20!$

Open solution

Q207. JEE Main 2022 (Online) 26th July Evening Shift · MCQ (Single Correct Answer)

$\sum_{\substack{i,j=0\\ i\neq j}}^{n} {}^nC_i\,{}^nC_j$ is equal to

(A) $2^{2n}-{}^{2n}C_n$ (B) $2^{2n-1}-{}^{2n-1}C_{n-1}$

(C) $2^{2n}-\frac{1}{2}{}^{2n}C_n$ (D) $2^{2n-1}+{}^{2n-1}C_n$

Open solution

Q208. JEE Main 2022 (Online) 29th June Evening Shift · MCQ (Single Correct Answer)

Let n $\geq$ 5 be an integer. If $9^n-8n-1=64\alpha$ and $6^n-5n-1=25\beta$, then $\alpha-\beta$ is equal to

(A) $1+{}^nC_2(8-5)+{}^nC_3(8^2-5^2)+\ldots..+{}^nC_n(8^{n-1}-5^{n-1})$

(B) $1+{}^nC_3(8-5)+{}^nC_4(8^2-5^2)+\ldots..+{}^nC_n(8^{n-2}-5^{n-2})$

(C) ${}^nC_3(8-5)+{}^nC_4(8^2-5^2)+\ldots..+{}^nC_n(8^{n-2}-5^{n-2})$

(D) ${}^nC_4(8-5)+{}^nC_5(8^2-5^2)+\ldots..+{}^nC_n(8^{n-3}-5^{n-3})$

Open solution

Q209. JEE Main 2022 (Online) 28th June Morning Shift · MCQ (Single Correct Answer)

If

$$\sum_{k=1}^{31}\left({}^{31}C_k\right)\left({}^{31}C_{k-1}\right)-\sum_{k=1}^{30}\left({}^{30}C_k\right)\left({}^{30}C_{k-1}\right)=\frac{\alpha(60!)}{(30!)(31!)},$$

where $\alpha\in$ R, then the value of 16α is equal to

(A) 1411 (B) 1320 (C) 1615 (D) 1855

Open solution

Q210. JEE Main 2021 (Online) 27th August Morning Shift · MCQ (Single Correct Answer)

$\sum_{k=0}^{20}\left({}^{20}C_k\right)^2$ is equal to :

(A) ${}^{40}C_{21}$ (B) ${}^{40}C_{19}$ (C) ${}^{40}C_{20}$ (D) ${}^{41}C_{20}$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q211. JEE Main 2021 (Online) 26th August Morning Shift · MCQ (Single Correct Answer)

If ${}^{20}C_r$ is the co-efficient of x^r in the expansion of $(1 + x)^{20}$, then the value of $\sum_{r=0}^{20} r^2 . {}^{20}C_r$ is equal to :

(A) 420×2^{19} (B) 380×2^{19} (C) 380×2^{18} (D) 420×2^{18}

Open solution

Q212. JEE Main 2021 (Online) 20th July Evening Shift · MCQ (Single Correct Answer)

For the natural numbers m, n, if $(1 - y)^m (1 + y)^n = 1 + a_1 y + a_2 y^2 + \ldots . + a_{m+n} y^{m+n}$ and $a_1 = a_2 = 10$, then the value of (m + n) is equal to :

(A) 88 (B) 64 (C) 100 (D) 80

Open solution

Q213. JEE Main 2021 (Online) 20th July Morning Shift · MCQ (Single Correct Answer)

The coefficient of x^{256} in the expansion of

$(1 - x)^{101} (x^2 + x + 1)^{100}$ is :

(A) ${}^{100}C_{16}$ (B) ${}^{100}C_{15}$ (C) $- {}^{100}C_{16}$ (D) $- {}^{100}C_{15}$

Open solution

Q214. JEE Main 2021 (Online) 18th March Morning Shift · MCQ (Single Correct Answer)

Let $(1 + x + 2x^2)^{20} = a_0 + a_1 x + a_2 x^2 + ... + a_{40} x^{40}$. Then $a_1 + a_3 + a_5 + ... + a_{37}$ is equal to

(A) $2^{20}(2^{20} - 21)$ (B) $2^{19}(2^{20} - 21)$ (C) $2^{19}(2^{20} + 21)$ (D) $2^{20}(2^{20} + 21)$

Open solution

Q215. JEE Main 2021 (Online) 17th March Evening Shift · MCQ (Single Correct Answer)

The value of $\sum_{r=0}^{6} \left({}^6C_r . {}^6C_{6-r}\right)$ is equal to :

(A) 924 (B) 1024 (C) 1124 (D) 1324

Open solution

Q216. JEE Main 2021 (Online) 16th March Morning Shift · MCQ (Single Correct Answer)

Let [x] denote greatest integer less than or equal to x. If for $n \in N$,

$(1 - x + x^3)^n = \sum_{j=0}^{3n} a_j x^j$,

then $\sum_{j=0}^{\left[\frac{3n}{2}\right]} a_{2j} + 4 \sum_{j=0}^{\left[\frac{3n-1}{2}\right]} a_{2j+1}$ is equal to :

(A) $2^n - 1$ (B) n (C) 2 (D) 1

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q217. JEE Main 2021 (Online) 24th February Evening Shift · MCQ (Single Correct Answer)

If $n \geq 2$ is a positive integer, then the sum of the series ${}^{n+1}C_2 + 2\left({}^2C_2 + {}^3C_2 + {}^4C_2 + \ldots + {}^nC_2\right)$ is :

(A) $\frac{n(2n+1)(3n+1)}{6}$ (B) $\frac{n(n+1)(2n+1)}{6}$

(C) $\frac{n(n+1)^2(n+2)}{12}$ (D) $\frac{n(n-1)(2n+1)}{6}$

Open solution

Q218. JEE Main 2021 (Online) 24th February Morning Shift · MCQ (Single Correct Answer)

The value of
$-{}^{15}C_1 + 2.{}^{15}C_2 - 3.{}^{15}C_3 + \ldots - 15.{}^{15}C_{15} + {}^{14}C_1 + {}^{14}C_3 + {}^{14}C_5 + \ldots + {}^{14}C_{11}$ is :

(A) $2^{13} - 13$ (B) $2^{16} - 1$ (C) 2^{14} (D) $2^{13} - 14$

Open solution

Q219. JEE Main 2020 (Online) 4th September Evening Slot · MCQ (Single Correct Answer)

If for some positive integer n, the coefficients of three consecutive terms in the binomial expansion of $(1 + x)^{n+5}$ are in the ratio 5 : 10 : 14, then the largest coefficient in this expansion is :

(A) 330 (B) 792 (C) 252 (D) 462

Open solution

Q220. JEE Main 2020 (Online) 4th September Morning Slot · MCQ (Single Correct Answer)

The value of $\sum_{r=0}^{20} {}^{50-r}C_6$ is equal to:

(A) ${}^{50}C_6 - {}^{30}C_6$ (B) ${}^{51}C_7 - {}^{30}C_7$ (C) ${}^{50}C_7 - {}^{30}C_7$ (D) ${}^{51}C_7 + {}^{30}C_7$

Open solution

Q221. JEE Main 2019 (Online) 12th April Evening Slot · MCQ (Single Correct Answer)

If ${}^{20}C_1 + (2^2)\,{}^{20}C_2 + (3^2)\,{}^{20}C_3 + \ldots\ldots + (20^2)\,{}^{20}C_{20} = A(2^{\beta})$, then the ordered pair (A, β) is equal to :

(A) (420, 19) (B) (420, 18) (C) (380, 18) (D) (380, 19)

Open solution

Q222. JEE Main 2019 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

If the coefficients of x^2 and x^3 are both zero, in the expansion of the expression $(1 + ax + bx^2)(1 - 3x)^{15}$ in powers of x, then the ordered pair (a,b) is equal to :

(A) (28, 861) (B) (28, 315) (C) (−21, 714) (D) (−54, 315)

Open solution

Q223. JEE Main 2019 (Online) 9th April Evening Slot · MCQ (Single Correct Answer)

If some three consecutive coefficients in the binomial expansion of $(x + 1)^n$ in powers of x are in the ratio 2 : 15 : 70, then the average of these three coefficients is :-

(A) 625 (B) 227 (C) 964 (D) 232

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q224. JEE Main 2019 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

The sum of the series

$2.{}^{20}C_0 + 5.{}^{20}C_1 + 8.{}^{20}C_2 + 11.{}^{20}C_3 + \ldots + 62.{}^{20}C_{20}$ is equal to :

(A) 2^{25} (B) 2^{24} (C) 2^{26} (D) 2^{23}

Open solution

Q225. JEE Main 2019 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

The sum of the co-efficients of all even degree terms in x in the expansion of $\left(x+\sqrt{x^3-1}\right)^6 + \left(x-\sqrt{x^3-1}\right)^6$, (x > 1) is equal to:

(A) 32 (B) 26 (C) 29 (D) 24

Open solution

Q226. JEE Main 2019 (Online) 11th January Evening Slot · MCQ (Single Correct Answer)

Let $(x+10)^{50} + (x-10)^{50} = a_0 + a_1x + a_2x^2 + \ldots + a_{50}x^{50}$, for all $x \in R$; then $\frac{a_2}{a_0}$ is equal to

(A) 12.25 (B) 12.75 (C) 12.00 (D) 12.50

Open solution

Q227. JEE Main 2019 (Online) 11th January Evening Slot · MCQ (Single Correct Answer)

Let $S_n = 1 + q + q^2 + \ldots + q^n$ and $T_n = 1 + \left(\frac{q+1}{2}\right) + \left(\frac{q+1}{2}\right)^2 + \ldots + \left(\frac{q+1}{2}\right)^n$ where q is a real number and $q \neq 1$. If ${}^{101}C_1 + {}^{101}C_2 \cdot S_1 + \ldots + {}^{101}C_{101} \cdot S_{100} = \alpha T_{100}$ then α is equal to

(A) 202 (B) 200 (C) 2^{100} (D) 2^{99}

Open solution

Q228. JEE Main 2019 (Online) 11th January Morning Slot · MCQ (Single Correct Answer)

The value of r for which ${}^{20}C_r\,{}^{20}C_0 + {}^{20}C_{r-1}\,{}^{20}C_1 + {}^{20}C_{r-2}\,{}^{20}C_2 + \ldots + {}^{20}C_0\,{}^{20}C_r$ is maximum, is

(A) 20 (B) 15 (C) 10 (D) 11

Open solution

Q229. JEE Main 2019 (Online) 10th January Morning Slot · MCQ (Single Correct Answer)

If $\sum_{i=1}^{20}\left(\frac{{}^{20}C_{i-1}}{{}^{20}C_i + {}^{20}C_{i-1}}\right)^3 = \frac{k}{21}$ then k is equal to

(A) 100 (B) 200 (C) 50 (D) 400

Open solution

Q230. JEE Main 2019 (Online) 9th January Evening Slot · MCQ (Single Correct Answer)

The coefficient of t^4 in the expansion of $\left(\frac{1-t^6}{1-t}\right)^3$ is :

(A) 14 (B) 15 (C) 10 (D) 12

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q231. JEE Main 2018 (Online) 16th April Morning Slot · MCQ (Single Correct Answer)

The coefficient of x^2 in the expansion of the product $(2-x^2).((1 + 2x + 3x^2)^6 + (1 - 4x^2)^6)$ is :

(A) 107 (B) 106 (C) 108 (D) 155

Open solution

Q232. JEE Main 2018 (Offline) · MCQ (Single Correct Answer)

The sum of the co-efficients of all odd degree terms in the expansion of

$\left(x+\sqrt{x^3-1}\right)^5+\left(x-\sqrt{x^3-1}\right)^5, (x>1)$ is

(A) 2 (B) -1 (C) 0 (D) 1

Open solution

Q233. JEE Main 2018 (Online) 15th April Evening Slot · MCQ (Single Correct Answer)

The coefficient of x^{10} in the expansion of $(1 + x)^2(1 + x^2)^3(1 + x^3)^4$ is equal to :

(A) 52 (B) 56 (C) 50 (D) 44

Open solution

Q234. JEE Main 2018 (Online) 15th April Morning Slot · MCQ (Single Correct Answer)

If n is the degree of the polynomial,

$$\left[\frac{2}{\sqrt{5x^3+1}-\sqrt{5x^3-1}}\right]^8+\left[\frac{2}{\sqrt{5x^3+1}+\sqrt{5x^3-1}}\right]^8$$

and m is the coefficient of x^n in it, then the ordered pair (n, m) is equal to :

(A) $(24, (10)^8)$ (B) $(8, 5(10)^4)$ (C) $(12, (20)^4)$ (D) $(12, 8(10)^4)$

Open solution

Q235. JEE Main 2017 (Offline) · MCQ (Single Correct Answer)

The value of $\left({}^{21}C_1-{}^{10}C_1\right)+\left({}^{21}C_2-{}^{10}C_2\right)+\left({}^{21}C_3-{}^{10}C_3\right)$ $\left({}^{21}C_4-{}^{10}C_4\right)+\ldots.+\left({}^{21}C_{10}-{}^{10}C_{10}\right)$ is

(A) $2^{21}-2^{10}$ (B) $2^{20}-2^9$ (C) $2^{20}-2^{10}$ (D) $2^{21}-2^{11}$

Open solution

Q236. JEE Main 2016 (Offline) · MCQ (Single Correct Answer)

If the number of terms in the expansion of $\left(1-\frac{2}{x}+\frac{4}{x^2}\right)^n$, $x \neq 0$, is 28, then the sum of the coefficients of all the terms in this expansion, is :

(A) 243 (B) 729 (C) 64 (D) 2187

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q237. JEE Main 2015 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

If the coefficients of the three successive terms in the binomial expansion of $(1 + x)^n$ are in the ratio 1 : 7 : 42, then the first of these terms in the expansion is :

(A) 6th (B) 7th (C) 8th (D) 9th

Open solution

Q238. JEE Main 2015 (Offline) · MCQ (Single Correct Answer)

The sum of coefficients of integral power of x in the binomial expansion $\left(1 - 2\sqrt{x}\right)^{50}$ is :

(A) $\frac{1}{2}\left(3^{50} - 1\right)$ (B) $\frac{1}{2}\left(2^{50} + 1\right)$

(C) $\frac{1}{2}\left(3^{50} + 1\right)$ (D) $\frac{1}{2}\left(3^{50}\right)$

Open solution

Q239. JEE Main 2014 (Online) 11th April Morning Slot · MCQ (Single Correct Answer)

The coefficient of x^{50} in the binomial expansion of

$(1 + x)^{1000} + x(1 + x)^{999} + x^2(1 + x)^{998} + \ldots\ldots + x^{1000}$ is :

(A) $\frac{(1000)!}{(50)!\ (950)!}$ (B) $\frac{(1000)!}{(49)!\ (951)!}$

(C) $\frac{(1001)!}{(51)!\ (950)!}$ (D) $\frac{(1001)!}{(50)!\ (951)!}$

Open solution

Q240. JEE Main 2014 (Offline) · MCQ (Single Correct Answer)

If the coefficints of x^3 and x^4 in the expansion of $(1 + ax + bx^2)(1 - 2x)^{18}$ in powers of x are both zero, then (a, b) is equal to:

(A) $\left(14, \frac{272}{3}\right)$ (B) $\left(16, \frac{272}{3}\right)$

(C) $\left(16, \frac{251}{3}\right)$ (D) $\left(14, \frac{251}{3}\right)$

Open solution

Q241. JEE Main 2013 (Online) 25th April Morning Slot · MCQ (Single Correct Answer)

If for positive integers r > 1, n > 2, the coefficient of the $(3r)^{th}$ and $(r+2)^{th}$ powers of x in the expansion of $(1 + x)^{2n}$ are equal, then n is equal to:

(A) 2r + 1 (B) 2r - 1 (C) 3r (D) r + 1

Open solution

Q242. AIEEE 2012 (Online) 7th May · MCQ (Single Correct Answer)

If $f(y) = 1 - (y - 1) + (y - 1)^2 - (y - 1)^3 + \ldots - (y - 1)^{17}$ then the coefficient of y^2 in it is

(A) ${}^{17}C_2$ (B) ${}^{17}C_3$ (C) ${}^{18}C_2$ (D) ${}^{18}C_3$

Open solution

Q243. AIEEE 2011 · MCQ (Single Correct Answer)

The coefficient of x^7 in the expansion of $\left(1 - x - x^2 + x^3\right)^6$ is

(A) -132 (B) -144 (C) 132 (D) 144

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q244. AIEEE 2010 · MCQ (Single Correct Answer)

Open solution

Let $s_1 = \sum_{j=1}^{10} j\,(j-1)^{10}C_j$,

$s_2 = \sum_{j=1}^{10} j.\,{}^{10}C_j$ and

$s_3 = \sum_{j=1}^{10} j^2.\,{}^{10}C_j.$

Statement-1 : $S_3 = 55 \times 2^9$.

Statement-2 : $S_1 = 90 \times 2^8$ and $S_2 = 10 \times 2^8$.

(A) Statement - 1 is true, Statement- 2 is true; Statement - 2 is not a correct explanation for Statement - 1.

(B) Statement - 1 is true, Statement-2 is false.

(C) Statement - 1 is false, Statement-2 is true.

(D) Statement - 1 is true, Statement-2 is true; Statement - 2 is a correct explanation for Statement - 1.

Q245. AIEEE 2008 · MCQ (Single Correct Answer)

Open solution

Statement - 1 : $\sum_{r=0}^{n} (r+1)\,{}^{n}C_r = (n+2)2^{n-1}$.

Statement - 2 : $\sum_{r=0}^{n} (r+1)\,{}^{n}C_r x^r = (1+x)^n + nx(1+x)^{n-1}$.

(A) Statement - 1 is false, Statement - 2 is true

(B) Statement - 1 is true, Statement - 2 is true; Statement - 2 is a correct explanation for Statement - 1

(C) Statement - 1 is true, Statement - 2 is true; Statement - 2 is not a correct explanation for Statement - 1

(D) Statement - 1 is true, Statement - 2 is false

Q246. AIEEE 2007 · MCQ (Single Correct Answer)

Open solution

The sum of the series ${}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \ldots\ldots - \ldots\ldots + {}^{20}C_{10}$ is

(A) 0 (B) ${}^{20}C_{10}$ (C) $-{}^{20}C_{10}$ (D) $\frac{1}{2}{}^{20}C_{10}$

Q247. AIEEE 2006 · MCQ (Single Correct Answer)

Open solution

For natural numbers m , n, if $(1-y)^m(1+y)^n = 1 + a_1y + a_2y^2 + \ldots\ldots\ldots$ and $a_1 = a_2 = 10$, then $(m,\ n)$ is

(A) $(20,\ 45)$ (B) $(35,\ 20)$ (C) $(45,\ 35)$ (D) $(35,\ 45)$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q248. AIEEE 2005 · MCQ (Single Correct Answer)

The value of ${}^{50}C_4 + \sum_{r=1}^{6} {}^{56-r}C_3$ is

(A) ${}^{55}C_4$ (B) ${}^{55}C_3$ (C) ${}^{56}C_3$ (D) ${}^{56}C_4$

Open solution

Q249. AIEEE 2004 · MCQ (Single Correct Answer)

If $S_n = \sum_{r=0}^{n} \frac{1}{{}^nC_r}$ and $t_n = \sum_{r=0}^{n} \frac{r}{{}^nC_r}$, then $\frac{t_n}{S_n}$ is equal to

(A) $\frac{2n-1}{2}$ (B) $\frac{1}{2}n - 1$ (C) n - 1 (D) $\frac{1}{2}n$

Open solution

Q250. AIEEE 2002 · MCQ (Single Correct Answer)

If the sum of the coefficients in the expansion of $(a+b)^n$ is 4096, then the greatest coefficient in the expansion is

(A) 1594 (B) 792 (C) 924 (D) 2924

Open solution

Q251. AIEEE 2002 · MCQ (Single Correct Answer)

r and n are positive integers $r > 1$, $n > 2$ and coefficient of $(r+2)^{th}$ term and $3r^{th}$ term in the expansion of $(1+x)^{2n}$ are equal, then n equals

(A) $3r$ (B) $3r+1$ (C) $2r$ (D) $2r+1$

Open solution

Q252. JEE Main 2023 (Online) 6th April Morning Shift · Numerical

The coefficient of x^{18} in the expansion of $\left(x^4 - \frac{1}{x^3}\right)^{15}$ is _____.

Open solution

Q253. JEE Main 2023 (Online) 24th January Morning Shift · Numerical

Suppose $\sum_{r=0}^{2023} r^2 \, {}^{2023}C_r = 2023 \times \alpha \times 2^{2022}$. Then the value of α is _____

Open solution

Q254. JEE Main 2022 (Online) 29th July Evening Shift · Numerical

If $\sum_{k=1}^{10} k^2 \left({}^{10}C_k\right)^2 = 22000L$, then L is equal to _____.

Open solution

Q255. JEE Main 2022 (Online) 28th July Evening Shift · Numerical

If $1 + (2 + {}^{49}C_1 + {}^{49}C_2 + \ldots + {}^{49}C_{49})({}^{50}C_2 + {}^{50}C_4 + \ldots + {}^{50}C_{50})$ is equal to $2^n \cdot m$, where m is odd, then $n + m$ is equal to _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q256. JEE Main 2022 (Online) 26th July Morning Shift · Numerical

If the coefficients of x and x^2 in the expansion of $(1+x)^p(1-x)^q, p, q \leq 15$, are -3 and -5 respectively, then the coefficient of x^3 is equal to _____.

Open solution

Q257. JEE Main 2022 (Online) 27th June Evening Shift · Numerical

If the sum of the coefficients of all the positive powers of x, in the Binomial expansion of $\left(x^n + \frac{2}{x^5}\right)^7$ is 939, then the sum of all the possible integral values of n is _____.

Open solution

Q258. JEE Main 2022 (Online) 26th June Evening Shift · Numerical

If $\left({}^{40}C_0\right) + \left({}^{41}C_1\right) + \left({}^{42}C_2\right) + \ldots\ldots + \left({}^{60}C_{20}\right) = \frac{m}{n}{}^{60}C_{20}$ m and n are coprime, then m + n is equal to _____.

Open solution

Q259. JEE Main 2022 (Online) 25th June Morning Shift · Numerical

Let C_r denote the binomial coefficient of x^r in the expansion of $(1+x)^{10}$. If for $\alpha, \beta \in R$, $C_1 + 3.2C_2 + 5.3C_3 + \ldots\ldots$ upto 10 terms $= \frac{\alpha \times 2^{11}}{2^\beta - 1}\left(C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \ldots\ldots \; upto\, 10\, terms\right)$ then the value of $\alpha + \beta$ is equal to _____.

Open solution

Q260. JEE Main 2021 (Online) 1st September Evening Shift · Numerical

If the sum of the coefficients in the expansion of $(x + y)^n$ is 4096, then the greatest coefficient in the expansion is _____.

Open solution

Q261. JEE Main 2021 (Online) 26th August Evening Shift · Numerical

Let $\binom{n}{k}$ denotes ${}^nC_k$ and $\begin{bmatrix} n \\ k \end{bmatrix} = \begin{cases} \binom{n}{k}, & if\, 0 \leq k \leq n \\ 0, & otherwise \end{cases}$

If $A_k = \sum_{i=0}^{9} \binom{9}{i} \begin{bmatrix} 12 \\ 12-k+i \end{bmatrix} + \sum_{i=0}^{8} \binom{8}{i} \begin{bmatrix} 13 \\ 13-k+i \end{bmatrix}$ and $A_4 - A_3 = 190$ p, then p is equal to :

Open solution

Q262. JEE Main 2021 (Online) 25th July Evening Shift · Numerical

Let $n \in N$ and [x] denote the greatest integer less than or equal to x. If the sum of (n + 1) terms ${}^nC_0, 3.{}^nC_1, 5.{}^nC_2, 7.{}^nC_3, \ldots\ldots$ is equal to $2^{100} \cdot 101$, then $2\left[\frac{n-1}{2}\right]$ is equal to _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q263. JEE Main 2021 (Online) 22th July Evening Shift · **Numerical**

The number of elements in the set {n ∈ {1, 2, 3, ..., 100} | $(11)^n > (10)^n + (9)^n$} is _____.

Open solution

Q264. JEE Main 2021 (Online) 18th March Evening Shift · **Numerical**

Let ${}^nC_r$ denote the binomial coefficient of x^r in the expansion of $(1 + x)^n$. If $\sum_{k=0}^{10} (2^2 + 3k)^{10} C_k = \alpha.3^{10} + \beta.2^{10}, \alpha, \beta \in R$, then $\alpha + \beta$ is equal to _____.

Open solution

Q265. JEE Main 2021 (Online) 16th March Evening Shift · **Numerical**

Let n be a positive integer. Let

$A = \sum_{k=0}^{n} (-1)^k {}^nC_k \left[\left(\frac{1}{2}\right)^k + \left(\frac{3}{4}\right)^k + \left(\frac{7}{8}\right)^k + \left(\frac{15}{16}\right)^k + \left(\frac{31}{32}\right)^k \right]$. If

$63A = 1 - \frac{1}{2^{30}}$, then n is equal to _____.

Open solution

Q266. JEE Main 2021 (Online) 26th February Morning Shift · **Numerical**

Let m, n∈N and gcd (2, n) = 1. If $30\binom{30}{0} + 29\binom{30}{1} + \ldots\ldots + 2\binom{30}{28} + 1\binom{30}{29} = n.2^m$, then n + m is equal to _____.

(Here $\binom{n}{k} = {}^nC_k$)

Open solution

Q267. JEE Main 2021 (Online) 24th February Evening Shift · **Numerical**

For integers n and r, let $\binom{n}{r} = \begin{cases} {}^nC_r, & if\, n \geq r \geq 0 \\ 0, & otherwise \end{cases}$ The maximum value of k for which the sum $\sum_{i=0}^{k} \binom{10}{i}\binom{15}{k-i} + \sum_{i=0}^{k+1} \binom{12}{i}\binom{13}{k+1-i}$ exists, is equal to _____.

Open solution

Q268. JEE Main 2020 (Online) 5th September Evening Slot · **Numerical**

The coefficient of x^4 in the expansion of $(1 + x + x^2 + x^3)^6$ in powers of x, is _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q269. JEE Main 2020 (Online) 2nd September Evening Slot · **Numerical**

For a positive integer n,
$\left(1+\frac{1}{x}\right)^n$ is expanded
in increasing powers of x. If three consecutive coefficients in this expansion are in the ratio,
2 : 5 : 12, then n is equal to_____.

Open solution

Q270. JEE Main 2020 (Online) 9th January Evening Slot · **Numerical**

If $C_r \equiv {}^{25}C_r$ and
$C_0 + 5.C_1 + 9.C_2 + ... + (101).C_{25} = 2^{25}.k$, then k is equal to _____.

Open solution

Q271. JEE Main 2020 (Online) 7th January Morning Slot · **Numerical**

If the sum of the coefficients of all even powers of x in the product
$(1 + x + x^2 + ...+ x^{2n})(1 - x + x^2 - x^3 + ... + x^{2n})$ is 61, then n is equal to _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Answer Key

MCQ (Single Correct Answer)

1. **C**	2. **D**	3. **B**	4. **A**	5. **B**
6. **C**	7. **C**	8. **D**	9. **C**	10. **B**
11. **B**	12. **A**	13. **B**	14. **D**	15. **B**
16. **D**	17. **B**	18. **D**	19. **B**	20. **B**
21. **B**	22. **B**	23. **A**	24. **D**	25. **C**
26. **A**	27. **B**	28. **D**	29. **B**	30. **C**
31. **C**	32. **D**	33. **B**	34. **B**	35. **C**
36. **D**	37. **C**	38. **D**	39. **B**	40. *Scan QR*
41. **B**	42. **D**	43. **D**	44. **B**	45. **C**
46. **A**	47. **A**	48. **D**	49. **D**	50. **A**
51. **D**	52. **C**	53. **D**	54. **D**	55. **B**
56. **C**	57. **B**	58. **A**	59. **D**	60. **C**
61. **A**	62. **D**	63. **C**	64. **D**	65. **B**
66. **C**	67. **A**	68. **D**	69. **D**	70. **C**
71. **D**	72. **D**	73. **A**	74. **C**	75. **D**
76. **D**	77. **B**	78. **A**	79. **C**	80. **C**
81. **B**	82. **B**	83. **B**	84. **D**	85. **B**
86. **C**	87. **C**	88. **D**	89. **C**	90. **B**
91. **C**	92. **A**	93. **D**	94. **A**	95. **C**
96. **A**	97. **D**	98. **A**	99. **A**	100. **D**
101. **A**	102. **A**	103. **B**	104. **A**	105. **B**
106. **B**	107. **A**	108. **C**	109. **D**	110. **B**
111. **D**	112. **C**	113. **D**	114. **A**	115. **A**
116. **A**	117. **B**	118. **D**	119. **D**	120. **C**
121. **C**	122. **C**	123. **B**	124. **D**	125. **C**
126. **A**	127. **D**			

Numerical

128. **30**	129. **21**	130. **32**	131. **63**	132. **34**
133. **239**	134. **5**	135. **612**	136. **465**	137. **2035**
138. **1**	139. **5**	140. **806**	141. **54**	142. **8**

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

143. **678**	144. **25**	145. **118**	146. **10**	147. **138**
148. **1**	149. **2041**	150. **0**	151. **12**	152. **36**
153. **2736**	154. **171**	155. **960**	156. **1275**	157. **31**
158. **4**	159. **1**	160. **29**	161. **5040**	162. **98**
163. **9**	164. **2**	165. **23**	166. **1120**	167. **1**
168. **7**	169. **1080**	170. **405**	171. **34**	172. **84**
173. **57**	174. **24**	175. **6006**	176. **5**	177. **2**
178. **5**	179. **83**	180. **4**	181. **315**	182. **55**
183. **15**	184. **55**	185. **1**	186. **210**	187. **8**
188. **21**	189. **210**	190. **4**	191. **4**	192. **1**
193. **45**	194. **13**	195. **8**	196. **615**	

Out of syllabus

197. **A**	198. **C**	199. **A**	200. **B**	201. **B**
202. **D**	203. **C**	204. **B**	205. **C**	206. **B**
207. **A**	208. **C**	209. **A**	210. **C**	211. **D**
212. **D**	213. **B**	214. **B**	215. **A**	216. **D**
217. **B**	218. **D**	219. **D**	220. **B**	221. **B**
222. **B**	223. **D**	224. **A**	225. **D**	226. **A**
227. **C**	228. **A**	229. **A**	230. **B**	231. **B**
232. **A**	233. **A**	234. **C**	235. **C**	236. **B**
237. **B**	238. **C**	239. **D**	240. **B**	241. **A**
242. **D**	243. **B**	244. **B**	245. **B**	246. **D**
247. **D**	248. **D**	249. **D**	250. **C**	251. **C**
252. **5005**	253. **1012**	254. **221**	255. **99**	256. **23**
257. **57**	258. **102**	259. **286**	260. **924**	261. **49**
262. **98**	263. **96**	264. **19**	265. **6**	266. **45**
267. **12**	268. **120**	269. **118**	270. **51**	271. **30**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

ExamGOAL
PYQ PDF

Your Exam. Our Responsibility.

From first chapter to exam day.

Practise with the questions your exam has already asked, then test yourself under real exam conditions.

PAST YEAR QUESTIONS

Every past paper, chapter-wise.

Previous year questions sorted the way you study, each with a full solution.

- Chapter-wise and year-wise
- Weightage and topic tags on every chapter
- Your progress on each one, at a glance

MOCK TESTS

Mock tests with a real exam feel.

Practise under the same pressure as the real paper, on your phone or on the big screen.

- Timed sections, just like exam day
- Mark questions for review and come back

ANALYSIS

Know your score. Know your rank.

A full report after every test, so you know exactly where you stand.

- Your percentile among everyone who took it
- Predicted real-exam percentile and All India Rank
- Subject breakdown and time analysis

PRACTICE

Solve. Check. Understand.

Learn from every question, not just the ones you get right.

- Instant answers as you practise
- Step-by-step explanations
- See how everyone else answered

NCERT Exemplar
With full solutions

Exam syllabus
Subject and chapter-wise

Bookmarks
Organised in groups

Your notes
On any question

ExamGOAL
LEARNING APP

ExamGOAL: Exam Preparation App

One app for every exam. Free trial for new users, on Android and the web.

GET IT ON Google Play

Scan to get the app

www.examgoal.com
Practice and mock tests

questions.examside.com
Free solutions

pdf.questions.examgoal.com
Free PYQ PDFs